AMENDMENT OF SOLICITATION/MODIFICATION OF CONTRACT

1. CONTRACT ID CODE: J

2. AMENDMENT/MODIFICATION NUMBER	3. EFFECTIVE DATE	4. REQUISITION/PURCHASE REQUISITION NUMBER	5. PROJECT NUMBER *(If applicable)*
0003	25 AUG 2025		112339

6. ISSUED BY CODE W912EP

W074 ENDIST JACKSNVLLE
KO CONTRACTING DIVISION, 701 SAN MARCO BLVD
JACKSONVILLE, FL 32207-0019
UNITED STATES
MEREDITH COAD, EMAIL: MEREDITH.C.COAD@USACE.ARMY.MIL TELEPHONE: 904-232-3724

7. ADMINISTERED BY *(If other than Item 6)* CODE

SCD: PAS:

8. NAME AND ADDRESS OF CONTRACTOR *(Number, street, county, State and ZIP Code)*

CODE | FACILITY CODE

(X)

[X] 9A. AMENDMENT OF SOLICITATION NUMBER
W912EP25BA009

9B. DATED *(SEE ITEM 11)*
18 JUL 2025

[ ] 10A. MODIFICATION OF CONTRACT/ORDER NUMBER

10B. DATED *(SEE ITEM 13)*

11. THIS ITEM ONLY APPLIES TO AMENDMENTS OF SOLICITATIONS

[X] The above numbered solicitation is amended as set forth in Item 14. The hour and date specified for receipt of Offers [X] is extended. [ ] is not extended.

Offers must acknowledge receipt of this amendment prior to the hour and date specified in the solicitation or as amended, by one of the following methods:
(a) By completing items 8 and 15, and returning 1 copies of the amendment; (b) By acknowledging receipt of this amendment on each copy of the offer submitted; or (c) By separate letter or electronic communication which includes a reference to the solicitation and amendment numbers. FAILURE OF YOUR ACKNOWLEDGMENT TO BE RECEIVED AT THE PLACE DESIGNATED FOR THE RECEIPT OF OFFERS PRIOR TO THE HOUR AND DATE SPECIFIED MAY RESULT IN REJECTION OF YOUR OFFER. If by virtue of this amendment you desire to change an offer already submitted, such change may be made by letter or electronic communication, provided each letter or electronic communication makes reference to the solicitation and this amendment, and is received prior to the opening hour and date specified.

12. ACCOUNTING AND APPROPRIATION DATA *(If required)*

SEE SECTION G - CONTRACT ADMINISTRATION DATA

13. THIS ITEM APPLIES ONLY TO MODIFICATIONS OF CONTRACTS/ORDERS. IT MODIFIES THE CONTRACT/ORDER NUMBER AS DESCRIBED IN ITEM 14.

CHECK ONE	
[ ]	A. THIS CHANGE ORDER IS ISSUED PURSUANT TO: *(Specify authority)* THE CHANGES SET FORTH IN ITEM 14 ARE MADE IN THE CONTRACT ORDER NUMBER IN ITEM 10A.
[ ]	B. THE ABOVE NUMBERED CONTRACT/ORDER IS MODIFIED TO REFLECT THE ADMINISTRATIVE CHANGES *(such as changes in paying office, appropriation data, etc.)* SET FORTH IN ITEM 14, PURSUANT TO THE AUTHORITY OF FAR 43.103(b).
[ ]	C. THIS SUPPLEMENTAL AGREEMENT IS ENTERED INTO PURSUANT TO AUTHORITY OF:
[ ]	D. OTHER *(Specify type of modification and authority)*

E. IMPORTANT: Contractor [ ] is not [ ] is required to sign this document and return ______ copies to the issuing office.

14. DESCRIPTION OF AMENDMENT/MODIFICATION (*Organized by UCF section headings, including solicitation/contract subject matter where feasible.)*

COASTAL STORM RISK MANAGEMENT PROJECT HUTCHINSON ISLAND, ST. LUCIE COUNTY, FLORIDA W912EP25BA009 AMENDMENT 0003

THE BID DUE DATE HAS BEEN POSTPONED TO SEPTEMBER 2, 2025 @ 1:00 PM EDT. ALL OTHER TERMS AND CONDITIONS REMAIN UNCHANGED.

Except as provided herein, all terms and conditions of the document referenced in Item 9A or 10A, as heretofore changed, remains unchanged and in full force and effect.

15A. NAME AND TITLE OF SIGNER *(Type or print)*		16A. NAME AND TITLE OF CONTRACTING OFFICER *(Type or print)*	
		GRISELLE GONZALEZ-AQUINO	
15B. CONTRACTOR/OFFEROR	**15C. DATE SIGNED**	**16B. UNITED STATES OF AMERICA**	**16C. DATE SIGNED**
(Signature of person authorized to sign)		Griselle Gonzalez Aquino *(Signature of Contracting Officer)*	25 AUG 2025

Previous edition unusable

SF 30 CONTINUATION SHEET

Coastal Storm Risk Management Project
South Hutchinson Island
St. Lucie County, Florida

SUMMARY OF CHANGES

1. SPECIFICATIONS:

Volume 2:

DELETE Section 01 57 20, and **REPLACE** with the attached revised Section 01 57 20.

Added Section 01 57 20 Appendix B.

2. DRAWINGS: No change to drawings.

3. For Information Only:

(End of Summary of Changes)

SECTION SF 30 BLOCK 14 CONTINUATION PAGE

SUMMARY OF CHANGES CONT.

Section 00 00 00 - Procurement and Contracting Requirements

The following changes have been made:

INFORMATION	FROM	TO
Contract Description	DESCRIPTION OF WORK: REFER TO SECTION 01 00 00 FOR COMPLETE PROJECT DESCRIPTION DRAWINGS: SEE DFARS CLAUSE 252.236-7001, SECTION 00 70 00. MAGNITUDE OF CONSTRUCTION IS BETWEEN $10,000,000.00 - $25,000,000.00. THIS PROCUREMENT IS UNRESTRICTED. THE NAICS CODE FOR THIS PROJECT IS 237990 AND THE SMALL BUSINESS SIZE STANDARD IS $37 MILLION. IN ACCORDANCE WITH (IAW) FAR CLAUSE 52.236-27, AN ORGANIZED SITE VISIT HAS BEEN SCHEDULED. SEE FAR 52.236-27, ALT I IN SECTION 00 20 00 FOR DETAILS. YOU MUST BE REGISTERED IN THE SYSTEM FOR AWARD MANAGEMENT (SAM) TO RECEIVE AN AWARD FROM THIS SOLICITATION. THE SAM WEBSITE IS LOCATED AT WWW.SAM.GOV. **SEE BLOCK 11 - SEE SECTION 00 70 00, FAR CLAUSE 52.211-10 FOR PERIOD OF PERFORMANCE. BID OPENING WILL BE HELD VIRTUALLY ~~26 AUGUST~~ 2025 AT 2:00 PM (EDT) VIA MICROSOFT TEAMS. UTILIZE THE INFORMATION PROVIDED BELOW TO JOIN THE TEAMS MEETING: HTTPS://WWW.MICROSOFT.COM/EN-US/MICROSOFT-TEAMS/JOIN-A-MEETING MEETING ID: 993 970 512 120 / PASSCODE: S5e3En7f; DIAL IN BY PHONE: +1 601-262-2433 / PHONE	DESCRIPTION OF WORK: REFER TO SECTION 01 00 00 FOR COMPLETE PROJECT DESCRIPTION DRAWINGS: SEE DFARS CLAUSE 252.236-7001, SECTION 00 70 00. MAGNITUDE OF CONSTRUCTION IS BETWEEN $10,000,000.00 - $25,000,000.00. THIS PROCUREMENT IS UNRESTRICTED. THE NAICS CODE FOR THIS PROJECT IS 237990 AND THE SMALL BUSINESS SIZE STANDARD IS $37 MILLION. IN ACCORDANCE WITH (IAW) FAR CLAUSE 52.236-27, AN ORGANIZED SITE VISIT HAS BEEN SCHEDULED. SEE FAR 52.236-27, ALT I IN SECTION 00 20 00 FOR DETAILS. YOU MUST BE REGISTERED IN THE SYSTEM FOR AWARD MANAGEMENT (SAM) TO RECEIVE AN AWARD FROM THIS SOLICITATION. THE SAM WEBSITE IS LOCATED AT WWW.SAM.GOV. **SEE BLOCK 11 - SEE SECTION 00 70 00, FAR CLAUSE 52.211-10 FOR PERIOD OF PERFORMANCE. BID OPENING WILL BE HELD VIRTUALLY 2 SEPTEMBER 2025 AT 2:00 PM (EDT) VIA MICROSOFT TEAMS. UTILIZE THE INFORMATION PROVIDED BELOW TO JOIN THE TEAMS MEETING: HTTPS://WWW.MICROSOFT.COM/EN-US/MICROSOFT-TEAMS/JOIN-A-MEETING MEETING ID: 993 970 512 120 / PASSCODE: S5e3En7f; DIAL IN BY PHONE: +1 601-262-2433 / PHONE

	CONFERENCE ID: 490 634 637# **SEE ELECTRONIC BIDS IN SECTION 00 21 00 FOR SUBMISSION INSTRUCTIONS A PRE-BID CONFERENCE WILL ALSO BE HELD VIRTUALLY 4 AUGUST 2025 AT 2:00 PM EDT THROUGH THE FOLLOWING MS TEAMS MEETING: MEETING ID: 993 743 890 063 / PASSCODE: Jy2of2bM; DIAL IN BY PHONE: +1 601-262-2433 / PHONE CONFERENCE ID: 709 065 634#	CONFERENCE ID: 490 634 637# **SEE ELECTRONIC BIDS IN SECTION 00 21 00 FOR SUBMISSION INSTRUCTIONS A PRE-BID CONFERENCE WILL ALSO BE HELD VIRTUALLY 4 AUGUST 2025 AT 2:00 PM EDT THROUGH THE FOLLOWING MS TEAMS MEETING: MEETING ID: 993 743 890 063 / PASSCODE: Jy2of2bM; DIAL IN BY PHONE: +1 601-262-2433 / PHONE CONFERENCE ID: 709 065 634#
Response Due Date	~~26 Aug 2025~~	02 Sep 2025

Section 00 21 00 - Instructions

Miscellaneous text in this section has been modified to:

ELECTRONIC BIDS

(a) Definition:
Electronic bid, as used in this provision, means a bid, revision or modification of a bid, or withdrawal of a bid that is transmitted to and received by the Government via email or other means for transmitting electronic data. No flash drives will be accepted.

(b) Offerors must submit electronic bids in responses to this solicitation in order to be considered for award. Electronic bids are subject to the same rules as paper bids.

(c) If any portion of bid received by the Contracting Officer is unreadable to the degree that conformance to the essential requirements of the solicitation cannot be ascertained from the document.

(1) The Contracting Officer shall immediately notify the offeror and permit the offeror to resubmit the bid if the bid has been received prior to due date indicated in the solicitation.

(2) The method and time for resubmission shall by prescribed by the Contracting Officer after consultation with the offeror.

(d) The Government reserves the right to make award solely on the electronic bid. However, if requested to do so by the Contracting Officer, the apparently successful offeror shall submit the complete original signed bid.

(e) The Offeror's bid shall be submitted electronically, as described below. The use of hyperlinks in the bid is prohibited. The offeror's bid is due no later than 1:00 PM (EDT) on ~~26 August~~ 2 September 2025.

NOTE: The only authorized transmission method of bids in response to the subject solicitation is electronically via the Procurement Integrated Enterprise Environment (PIEE) website. No other transmission methods, such as email, facsimile, hand-carried, proprietary or DOD SAFE submissions, bids submitted by mail, etc. will be accepted. Offers submitted via any other transmission method other than the Procurement Integrated Enterprise Environment (PIEE) website will not be evaluated.

The Offeror's bid must be received by the Government no later than the date/time specified in the solicitation and cited above. Offerors shall submit their bid using the following link: https://piee.eb.mil/

Instructions for submitting electronic bids:

In an effort to reduce paperwork and cost, all bids shall be submitted electronically through the Procurement Integrated Enterprise Environment (PIEE) website.

Electronic copies of each volume shall be submitted through the Solicitation Module of PIEE at https://piee.eb.mil. If there is a system outage, see below. All bids received after the exact time specified for receipt shall be treated as late submissions.

For instructions on how to post an offer, please refer to the Posting Offer demo: https://pieetraining.eb.mil/wbt/sol/Posting_Offer.pdf.

It is the Offeror's responsibility to obtain written confirmation of receipt of all electronic files of the full bid by the Contracting office. The submission date for all Volumes shall be no later than the date and time specified in Block 13 of the SF 1442 of the Solicitation. In the unlikely event the PIEE system is not operational, experiences technical difficulties, or a Contractor is temporarily unable to access or use the system, the Contractor shall immediately notify the Contracting Officer. This Notification must occur prior to the bid submission deadline. Contractor notification shall be in writing and may be in conjunction with verbal notification, but verbal notification alone shall not be sufficient.

The alternate method for bid submission is via expedited delivery or hand-carried. The Offeror must obtain prior approval from the Contracting Officer to use the alternate submission method. The Contractor shall have 24 hours to provide their bid upon contracting officer approval. In the event that the Contractor alleges technical difficulties and does not notify the contracting officer until after the closing date and time, the contracting officer
will follow the procedures identified in the Federal Acquisition Regulation (FAR) and then make a determination if the Contractor's late bid submission is accepted.

Screen shots of the submission should be taken to validate a submission was accepted in the PIEE system. You are limited to five (5) maximum files per upload (total size cannot exceed 2GB). If you have a large number of files, it is recommended that you combine or ZIP your files before uploading to the PIEE site. Offerors may use compression utility software such as WinZip or PKZip to reduce file size and facilitate transmission.

File Description:

To ensure your submission is received and processed appropriately, it is important that interested parties CAREFULLY ensure their electronic files adhere to the following naming convention. Each file name shall begin with the solicitation number, followed by the word "RESPONSE", followed by your firm's name and finally a brief file description.

EXAMPLE:
"W912EP25BA009 RESPONSE FIRM NAME.pdf"

File Organization, Formatting, and other instructions:

Each file shall be clearly indexed and logically assembled. Font size shall be 10 or larger. Pages shall be letter sized--larger page sizes (such as 11x17 foldouts, etc.) will be counted as two pages. Offerors shall prepare bids in the English language. Bids shall be in a narrative format, organized and titled so that each section of the bid follows the order and format of the factors. Information presented should be organized so as to pertain to only the evaluation factor in the section that the information is presented. Information pertaining to more than one evaluation factor should be repeated in each section for each factor.

Include a "File Description" for each file(s) you upload. The "File Description" will be included in the email notice to each of the recipients you choose to have access your file(s).

NOTE: Do NOT enter Privacy Act Data (Personal Identification Information (PII)) in the File Description.)

Upload Completion & Deadline:

Interested parties shall submit responses no later than the date cited above. Offerors should time their upload effort with prudence by not waiting until the last few minutes--this will allow for unexpected delays in the transmittal process. Submissions after the deadline may be emailed to Contract Specialist Meredith Coad at (Meredith.C.Coad@usace.army.mil) but be advised, they will be considered late and as such, will be processed in accordance with FAR 15.208.

Submission shall be in Adobe PDF format.

Receipt of Submission:

For the purposes of establishing whether a bid submission is considered timely, the Government considers the date and time the submission is completely uploaded into the PIEE website. For multiple submissions, the Government will consider the date and time the last submission is completely uploaded into the PIEE website. Do not assume that electronic communication is instantaneous. It can take several minutes or even hours in some cases. The Government will not be responsible for submissions delivered to any location or to anyone other than those designated to receive bids. Offerors are responsible for ensuring that offers are submitted so as to reach the designated recipient. Offerors are responsible for allowing sufficient time for the offer to be received in accordance with the instructions provided.

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 57 20

ENVIRONMENTAL PROTECTION

-- End of Section Table of Contents --

SECTION 01 57 20

ENVIRONMENTAL PROTECTION

PART 1 GENERAL

1.1 SCOPE

This Section covers prevention of environmental damage as the result of construction operations under this contract and for those measures set forth in other Technical Requirements of these specifications. For the purpose of this specification, environmental damage is defined as the presence of hazardous, physical, chemical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances; affect other species, biological communities, or ecosystems; or degrade the quality of the environment for aesthetic, cultural, and/or historical purposes. The control of environmental damage requires consideration of land, water, and air, and includes management of visual aesthetics, noise, solid waste, radiant energy and radioactive materials, as well as other pollutants.

1.2 ENVIRONMENTAL LAWS, REGULATIONS AND DOCUMENTS

1.2.1 Miscellaneous Environmental Laws And Regulations

There are numerous environmental laws and regulations. At the Federal level, the applicable laws and regulations include compliance with the Clean Water Act (CWA); Clean Air Act (CAA); Coastal Zone Management Act (CZMA); Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA); Endangered Species Act (ESA); Fish and Wildlife Coordination Act (FWCA); Marine Protection, Research, and Sanctuaries Act (MPRSA); Magnuson-Stevens Fishery Conservation and Management Act (MSFCMA); National Environmental Policy Act (NEPA); National Historic Preservation Act (NHPA); National Pollutant Discharge Elimination System (NPDES); Research and Sanctuaries Act (RSA); Native American Graves Protection and Repatriation Act (NAGPRA); Resource Conservation and Recovery Act (RCRA); Rivers and Harbors Act (R&H); Safe Drinking Water Act (SDWA); Toxic Substance Control Act (TSCA); Wild and Scenic Rivers Act (WSRA); Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA); Marine Mammal Protection Act (MMPA); Migratory Bird Treaty Act (MBTA); Code of Federal Regulations (CFRs); Executive Orders (EO); and, Environmental Protection Agency (EPA) requirements.

1.2.2 NEPA Reference

Commitments for the Contractor made in the NEPA document have been incorporated into these specifications. For additional background information, the NEPA document can be found at the following website: https://www.saj.usace.army.mil/About/Divisions-Offices/Planning/Environmental-Branch/Environmental-Documents/

1.3 REFERENCES

Any publication listed below forms a part of this specification to the extent referenced. Publications are referred to in the text by basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety and Occupational Health (SOH) Requirements
EM 1110-1-1003	(2013) NAVSTAR Global Positioning System Surveying

ILLUMINATING ENGINEERING SOCIETY OF NORTH AMERICA (IES)

IES RP-12	(1997) Recommended Practice for Marine Lighting
IES RP-16	(2010; Addendum A 2008; Addenda B & C 2009) Nomenclature and Definitions for Illuminating Engineering

Code of Federal Regulations (CFR) indicated herein may be found at https://www.govinfo.gov/app/collection/cfr.

1.4 QUALITY CONTROL

Establish and maintain quality control for environmental protection of all items set forth herein. Record on daily quality control reports or attachments thereto, any problems in complying with laws, regulations and ordinances, and corrective action taken.

1.5 PERSONNEL TRAINING

The Contractor is to train their personnel in all phases of environmental protection.

1.6 PERMITS AND AUTHORIZATIONS

Comply with all requirements under the terms and conditions set out in the following permit(s) and authorization(s) obtained by the Corps of Engineers listed below.

a. Florida Department of Environmental Protection Permit No. 0154626-001-JC; Effective Date: June 6, 2012 and modification 0154626-008-JN; Effective Date: July 30, 2021; Expiration Date: June 6, 2027; and modification 0154626-010-JN Effective Date: March 10, 2025; Expiration Date: June 6, 2027.

b. Statewide Programmatic Biological Opinion. Shore Protection Activities along the Coast of Florida. Service Log Number 41910-2011-F-0170. February 27, 2015.

c. Bureau of Ocean Energy Management (BOEM) Negotiated Lease Agreement No. OCS-A 0548 issuedAugust 14, 2025.

d. 2020 South Atlantic Regional Biological Opinion for Dredging and Material Placement Activities in the Southeast United States. ECO Tracking Number SERO-2019-03111. July 30, 2020.

e. 2013 Programmatic Piping Plover Biological Opinion for the effects of U.S. Army Corps of Engineers (Corps) planning and regulatory shore protection activities on the non-breeding piping plover (Charadrius melodus) and its designated Critical Habitat. Consultation Code O4EF1000-2013-F-0124. May 22, 2013.

The above permits indicate issuing agency approval of work required by this contract. Permits are available at the following web address: https://www.saj.usace.army.mil/About/Divisions-Offices/Planning/Environmental-Branch/Environmental-Compliance/

Obtain and comply with the requirements of all other permits or licenses required for construction of this project in accordance with the Clause PERMITS AND RESPONSIBILITIES of Section 00700 CONTRACT CLAUSES in Volume 1.

1.7 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Environmental Protection Plan; G, DO

Pre-Construction Pipeline Corridor Hardbottom Survey Report; G, DO

Winter Shorebird Survey; G, DO

Pre-Construction Pipeline Corridor Hardbottom Survey; G, DO

SD-02 Shop Drawings

Turtle Deflector Device; G, DO

SD-06 Test Reports

Escarpment Site Visit Report(s); G, DO

Post-Placement Pre-Pumping Pipeline Survey Results; G, DO

Log Of The Results Of Turtle Egg Monitoring And Recovery Activities; G, DO

Protected Listed Species Sighting Reports; G, DO

Weekly Logs/Summary Of FWC Hopper Dredge Reports; G, DO

Weekly Protected Species Summary Report; G, DO

SD-07 Certificates

Marine Turtle Nesting Monitor And Relocation Qualifications; G, DO

Bird Monitoring Qualifications; G, DO

Marine Animal Observer Qualifications (Clamshell/Mechanical Only); G, DO

Endangered Species Observer and Trawl Supervisor Qualifications

(For Hopper Dredges Only); G, DO

Scientific Diver(S) Qualifications; G, DO

SD-11 Closeout Submittals

Protected Species Summary Report; G, DO

Logs/Summary Of Bird Nesting Monitoring Report; G, DO

Logs/Summary Of Marine Turtle Nesting And Relocation; G, DO

Comprehensive Trawling Daily Logs; G, DO

Invasive And Nuisance Species Report; G, DO

Project Environmental Summary Sheet; G, DO

Post-Construction Pipeline Corridor Hardbottom Survey; G, DO

1.8 ENVIRONMENTAL PROTECTION PLAN

Submit a draft Environmental Protection Plan no later than 5 calendar days after Notice to Proceed for review and acceptance by the Contracting Officer and cc the Environmental Compliance POC. The Government will consider an interim plan for the first 30 days of operations. However, furnish an acceptable final plan no later than 30 calendar days after receipt of Notice to Proceed. Acceptance of the plan will not relieve the Contractor of their responsibility for adequate and continuing control of pollutants and other environmental protection measures. Acceptance of the plan is conditional and predicated on satisfactory performance during construction. The Government reserves the right to require the Contractor to make changes to the Environmental Protection Plan or operations if the Contracting Officer determines that environmental protection requirements are not being met. No physical work at the site is to begin prior to acceptance of the plan or an interim plan covering the work to be performed.

The EPP is to have a Table of Contents. Pages of the EPP will be numbered, including appendices. Sections of the plan and all appendices are to be indexed in the table of contents. The Environmental Protection Plan is to include but not be limited to the following:

a. A list of Federal, State, and local laws, regulations, and permits concerning environmental protection, pollution control, and abatement that are applicable to the proposed operations and the requirements imposed by those laws, regulations, and permits.

b. Methods for protection of features to be preserved within authorized work areas. Prepare a listing of methods to protect resources needing protection, i.e., trees, shrubs, vines, grasses and ground cover, landscape features, air and water quality, fish and wildlife, soil, historical, archeological, and cultural resources.

c. Procedures to be implemented to provide the required environmental protection and to comply with the applicable laws and regulations. Provide written assurance that immediate corrective action will be taken to correct pollution of the environment due to accident, natural causes, or failure to follow the procedures set out in accordance with the environmental protection plan.

d. A permit or license for and the location of the solid waste disposal area.

e. Drawings showing locations of any proposed temporary excavations or embankments for haul roads, stream crossing, material storage areas, structures, sanitary facilities, and stockpiles of excess or spoil materials.

f. Environmental monitoring plans for the job site, including land, water, air, and noise monitoring.

g. Traffic control plan.

h. Methods of protecting surface and ground water during construction activities.

i. Spill prevention. Specify all potentially hazardous substances to be used on the job site and intended actions to prevent accidental or intentional introduction of such materials into the air, ground, water, wetlands, or drainage areas. Specify the provisions to be taken to meet Federal, State, and local laws and regulations regarding labeling, storage, removal, transport, and disposal of potentially hazardous substances. List the Corps environmental POC's with phone numbers and email addresses. Specify that the contractor will notify the Corps Environmental Branch POC's by phone the same day any spill if possible or no later than the morning of the next day(to include weekends).

j. Spill contingency plan for hazardous, toxic, or petroleum material. There must be a secondary containment (110% of full tank capacity) for any fuel storage tanks (including pumps/generators). This is to include temporary fuel storage above 550 gallons. Any fuel storage container with more than 550 gallon capacity, trailer mounted or otherwise, left on the job site for more than 24 hours must comply with the secondary containment with over head cover unless specifically authorized by the COR. Fuel storage tanks must have over head cover to prevent rainwater from entering the secondary containment. The spill response material quantity must be sufficient to absorb/contain the maximum expected fuel storage.

k. Work area plan showing the proposed activity in each portion of the area and identifying the areas of limited use or nonuse. Include measures for marking the limits of use areas.

l. Plan of borrow area(s).

m. Include a statement as to the person who will be responsible for implementation of the Environmental Protection Plan. The Contractor personnel responsible is to report directly to the Contractor's top management and is to have the authority to act for the Contractor in all environmental protection matters.

n. Recycling and Waste Management Plan. Executive Order 12873 of 20 October 1993 requires a number of considerations in planning a project. Fallen trees should not be burned or buried to the maximum extent practical. Mulching, composting, and other uses for trees should be considered. Also, recovery of metals at the job site, including aluminum cans, should be considered with proceeds to be retained by the Contractor. Non-Federal recycling and waste minimization efforts are to also be incorporated into this plan.

o. Operational plan to achieve protection of sea turtles during hopper dredge(s) operation.

p. Invasive and Nuisance Species Transfer Prevention Plan. The invasive and nuisance species transfer prevention plan is to identify specific transfer prevention procedures and designated cleaning sites/locations. The Contractor is to provide a report with photographs verifying that equipment brought on the site was cleaned. The prevention plan and report are to be submitted to the Invasive Species Management Branch, Operations Division.

1.9 QUALIFICATIONS

The individual(s) are to have a combination of education and experience that demonstrates they have the knowledge and background to identify the species of concern and be able to fulfill the requirements of the environmental specification. Resume/qualification submittals must specifically address the knowledge/experience with the species of concern. Any specific requirements for an environmental observer/monitor called out in environmental coordination or authorization documents (such as agency coordination letters, biological opinions, or permits) regarding the project area must be also addressed in the submittals.

1.9.1 Marine Turtle Nesting Monitor and Relocation Qualifications

Submit to the Contracting Officer and cc the Environmental Compliance POC qualifications and Florida Fish and Wildlife Conservation Commission (FF&WCC)Marine Turtle no later than 10 calendar days after Notice to Proceed. A FF&WCC permitted subcontractor is required to be approved by the Contracting Officer to accomplish the sea turtle monitoring and relocation of this section unless they demonstrate to the satisfaction of the Contracting Officer the capability to accomplish sea turtle monitoring and relocation by obtaining a permit from the FF&WCC for daily monitoring and relocation that is specific to the project beach area.

1.9.2 Bird Monitoring Qualifications

No later than 10 calendar days after Notice to Proceed, the Contractor is to furnish to the Contracting Officer and cc the Environmental Compliance POC for approval, the qualifications of the bird monitor(s). Appropriate qualifications for bird monitor(s) are to have demonstrated ability to find and/or identify bird species, nesting behavior, eggs and nests, and habitat requirements. Bird monitor(s) are to be familiar with the general information, data collection protocol, and data entry procedures outlined on the FF&WCC Florida Shorebird Database website (http://www.flshorebirddatabase.org/). Consult with and coordinate all monitoring plans and activities with the Contracting Officer. The Bird Monitors will meet the following minimum qualifications:

a. Has previously participated in beach-nesting bird surveys of Florida beach nesting birds and relevant work experience (provide references or resume). Experience with previous projects must document the ability to do the following:

(1) identify all species of beach-nesting birds by sight and sound,

(2) identify breeding/territorial behaviors, and find nests of shorebirds that occur in the project area,

(3) identify habitats preferred by shorebirds nesting in the project area.

b. Have a clear working knowledge of, and adhere to, the Breeding Bird Protocol for Florida’s Seabirds and Shorebirds: (https://app.myfwc.com/crossdoi/shorebirds/)

c. Have completed full-length webinars: Route Surveyor Training and Rooftop Monitoring Training, including annual refresher training. Training resources can be found on the Florida Shorebird Database (FSD): https://app.myfwc.com/crossdoi/shorebirds/

d. Familiar with FF&WCC beach driving guidelines: www.myfwc.com/conservation/you-conserve/wildlife/beach-driving.

e. Experience posting beach-nesting bird sites, consistent with Florida Shorebird Alliance (FSA) Guidelines (https://flshorebirdalliance.org/resources/instructions-resources/).

f. Has registered as a contributor to the Florida Shorebird Database (FSD).

1.9.3 Marine Animal Observer Qualifications (Clamshell/Mechanical Only)

Submit to the Contracting Officer and cc the Environmental Compliance POC for approval, the qualifications of the marine animal observer no later than 10 calendar days after Notice to Proceed. A resume of demonstrated experience monitoring manatees and marine turtles and their behaviors in association with in-water construction projects is required. Specifically, include a table with the names and locations of specific in-water construction projects that observers have worked on. Also, the table is to include for each in-water construction project the permit number(s), the estimated amount of time (hours) spent observing, the estimated number of manatees and marine turtles seen, and whether the work was ever halted due to manatees or marine turtles in the area. Complete the FWC Protected Marine Species Observer Information Form (see CONSTRUCTION FORMS AND DETAILS) for each marine animal observer to document their experience.

1.9.4 Endangered Species Observer and Trawl Supervisor Qualifications (For Hopper Dredges Only)

Submit to the Contracting Officer and cc the Environmental Compliance POC the names of the endangered species observer-biologists (ESOs) to perform the monitoring for this contract no later than 10 calendar days after Notice to Proceed. Only NMFS pre-approved ESOs are authorized to monitor threatened/endangered species takes by hopper dredges. A pre-approved list of ESOs can be obtained by contacting NOAA Fisheries'Southeast Region, Protected Resources Division at (727) 824-5312.

1.9.5 Scientific Diver(S) Qualifications

St. Lucie County Only: Submit the names and qualifications of the individuals performing hardbottom monitoring for the purpose of pipeline placement to the Contracting Officer for review and approval no later than

10 calendar days after Notice to Proceed. These individuals will be certified SCUBA divers in accordance with Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS and EM 385-1-1 Section 10 and have a BS degree or higher in the study of marine biology or a comparable field, must have scientific knowledge of local benthic marine hardbottom habitats and their flora and fauna, and have professional experience in conducting hardbottom monitoring surveys. If additional monitoring team(s) are subcontracted, or new staff are added to the monitoring team, submit proposed changes and qualifications to the Contracting Officer for review at least 10 days prior to a monitoring event. The Contractor's selected biological monitoring firm is fully responsible for training of new staff members and subcontractors on the required monitoring procedures, as well as the QA/QC verification of their work.

1.10 SUBCONTRACTORS

Assurance of compliance with this section by subcontractors will be the responsibility of the Contractor.

1.11 NOTIFICATION

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with the aforementioned Federal, State, or local laws or regulations, permits and other elements of the environmental protection specifications. After receipt of such notice, inform the Contracting Officer of proposed corrective action and take such action as may be approved. If the Contractor fails to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No time extensions are to be granted or costs or damages allowed to the Contractor for any such suspension.

Within 30 calendar days following project completion, the Contractor is to submit to the Contracting Officer and cc the Environmental Compliance POC the Project Environmental Summary Sheet. The Project Environmental Summary Sheet is to provide absence or occurrence of environmental incidents, as indicated in paragraph CONSTRUCTION FORMS AND DETAILS below.

1.12 CONTRACTOR PERSONNEL QUALIFICATIONS IN POLLUTION CONTROL

Provide qualified personnel to perform all phases of environmental protection, including methods of detecting and avoiding pollution, familiarization with pollution standards, both statutory and contractual, and careful installation and monitoring of the project to ensure adequate and continuous environmental pollution control. Quality Control and supervisory personnel are to be thoroughly knowledgeable of Federal, State, and local laws, regulations, and permits as listed in the Environmental Protection Plan submitted by the Contractor. Identify the Quality Control personnel in the Quality Control Plan submitted in accordance with Section 01 45 05 DREDGING/BEACH FILL PLACEMENT - CONTRACTOR QUALITY CONTROL.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 PROTECTION OF ENVIRONMENTAL RESOURCES

For contract work, comply with all applicable Federal, State, or local laws and regulations. The environmental resources within the project boundaries and those affected outside the limits of permanent work under this Contract will be protected at least during the entire period of this Contract. Confine activities to areas defined by the drawings and specifications. Environmental protection is to be as stated in the following subparagraphs.

3.1.1 General Project Environmental Design and Installation Criteria

Some project sites have features that must not be impacted in any way, including cultural, historic, or archeological features. At all sites, minimize disturbance to existing features at the site to the extent possible, including vegetative, topographic, and drainage pattern features. Wetland, seagrass or hardbottoms impacts (temporary access, detours, staging areas, and other work area impacts) to project sites should be avoided and may require separate permitting action. Any wetlands temporarily impacted are to have its soil restored upon project completion. Expansion of previously permitted project footprints may likewise require separate permitting action.

In all cases, the installation of temporary systems to aid construction will provide for protection of the environment during handling, installing, storing, utilizing, transporting, servicing, testing, refilling, transferring, pumping, processing, removing waste products, repairing and maintaining systems and their components. Necessary protection is to also be considered that would prevent contamination of the environment from impacts to the systems caused by storm water runoff and flooding. Retrofit of temporary systems on project sites to modern environmental protection design standards will also be considered.

In the event environmental protection measures fail, implement procedures to control and correct environmental damage. Coordinate with the Contracting Officer if environmental protection measures fail.

3.1.1.1 Sewage-Based Systems Environmental Design and Installation Criteria

In general, there will be no waste or debris discharges of any kind for a project unless authorized by the Contracting Officer. This is to include the Contractor's providing sufficient temporary sanitary equipment and facilities for the project. The design and/or installation of temporary or permanent sewage systems are to ensure that waters will be free of effects of sewage discharges. Adhere to Federal, State, or local codes and requirements regarding sewage in the design, such as those of the EPA and, in the case of the State, Chapter 62-620 (Wastewater Facilities) of the FAC. Adhere to Best Management Practices from the applicable agencies in the design.

3.1.2 Protection of Land Resources

Prior to the beginning of any construction, identify all land resources to be preserved or avoided within the work area. Remove materials displaced into not cleared areas. Do not remove, cut, deface, injure, or destroy land resources including trees, shrubs, vines, grasses, topsoil, and land forms without special permission from the Contracting Officer. Engage a qualified tree surgeon to perform all tree surgery. Repair injuries to bark, trunk, branches, and roots of protected trees by dressing, cutting, and painting as specified for Class I Fine Pruning, of the National Arborist Association Pruning Standards for Shade Tree or as per State's Agricultural Extension Agency Guidelines, immediately as occurrences arise. No ropes, cables, or guys are to be fastened to or attached to any trees for anchorage unless specifically authorized. Where such special emergency use is permitted, provide effective protection for land and vegetation resources at all times as defined in the following subparagraphs.

3.1.2.1 Work Area Limits

Prior to any construction, mark the areas that are not required to accomplish all work to be performed under this Contract. Mark or fence isolated areas within the general work area to be saved and protected. Protect existing trees designated to remain. Provide protection of tree roots against noxious materials in solution caused by run-off or spillage. Fires are to be located outside the canopy of protected trees. Do not store materials, trailers, or equipment within the drip line of any protected tree. Protect monuments and markers before construction operations commence. Where construction operations are to be conducted during darkness, the markers will be visible. Convey to personnel the purpose of marking and/or protection of all necessary objects.

3.1.2.2 Contractor Facilities and Other Work Areas

Locate facilities, staging and other work areas as indicated in Section 01 50 02 TEMPORARY CONSTRUCTION FACILITIES, and in compliance with permits listed in the paragraph PERMITS AND AUTHORIZATIONS above and obtained for performance of work. Manage borrow or dredge material management (DMMA) areas to minimize erosion and to prevent sediment from entering nearby watercourses, wetlands, or lakes. Manage and control dredged material areas to limit spoil intrusion into areas designated on the drawings and to prevent erosion of soil or sediment from entering nearby watercourses, wetlands, or lakes. Develop dredged material areas in accordance with the grading plan indicated on the drawings. Control temporary excavation and embankments for plant and/or work areas to protect adjacent areas from despoilment. If there is suspicion that sediment may be unsuitable for disposal at a specified location, immediately take measures to contain the suspect sediment and notify the Contracting Officer.

3.1.2.3 Solid Wastes

The Contracting Officer will approve the type of solid waste containers, staging areas on land for these containers, the roads to haul the solid waste, and the final solid waste disposal site. In addition, comply with Federal, State and local regulations pertaining to the use of the solid waste disposal site.

Place solid wastes (excluding clearing debris) in containers which are emptied on a regular schedule. Conduct handling and disposal to prevent contamination.

3.1.2.4 Fuel, Oil, and Lubricants

Manage fuel, oil, and lubricants in order to prevent spills and evaporation. Provide a 4-foot square, 16-gauge metal pan with borders banded up and welded at corners right below the bibb for fuel dispensers to prevent spills. Edges of the pans are to be 8-inch minimum in depth to ascertain that no contamination of the ground takes place. Clean pans by an approved method immediately after every dispensing of fuel and wastes disposed of offsite in an approved area. If any spilling of fuel occurs, immediately recover the contaminated ground and dispose of it offsite in an approved area. Store petroleum waste generated in marked corrosion-resistant containers and recycled or disposed of in accordance with 40 CFR 279, State, and local regulations. Any fueling operation is to be conducted such that any accidental spills will be prevented from entering surface waters or wetlands.

For any fuel stored onsite overnight at the project work site in excess of 550 gallons, whether attached to a tractor trailer or not, the fuel storage area/fuel dispensing area is to be located within a bermed and if practical impervious lined area. If directed by the Contracting Officer, at no additional cost to the government, a fuel storage system with less than 550 gallons may be required to be within a bermed area. The bermed area must be sized to be able to contain the maximum amount of fuel stored on site and be sloped to allow access/egress to the vehicles being fueled at this location. Fueling operations, to the maximum extent practical, are to be conducted within this bermed/lined fuel storage area. These preventive measures to contain potential spills must be taken by the contractor at no additional cost to the government. If the construction activity is close to an OFW, the fuel storage bermed area must be lined unless this is not practical The COR may determine if other measures are necessary to prevent any fuel spills from entering nearby surface water/wetlands, and these additional actions will be required of the contractor at no additional cost to the government. Any fueling operation must be manned 100% of the time to ensure a rapid response to any potential fuel spillage. The fuel storage container is to be protected from vandalism by appropriate security measures which could include a fenced area with a lockable gate with high security locks on the gate and/or high security locks on the fuel dispensing equipment. These security measures are to be provided by the contractor at no additional cost to the government. The appropriate level of security measures are to be determined by the COR as appropriate to the individual job location. For fuel stored at the job site in tanks less than 550 gallons, the COR may determine a secondary containment/and or double walled tank be required at no additional cost to the government.

3.1.2.5 Hazardous Waste

Hazardous wastes are defined in 40 CFR 261. Ensure that hazardous wastes are stored and disposed of in accordance with 40 CFR 261 and State and local regulations. Ensure that hazardous wastes are packed, labeled, and transported in accordance with 49 CFR 173 and State and local regulations.

3.1.2.6 Hazardous Materials

Ensure that hazardous materials are labeled, stored, and transported in accordance with 49 CFR 173, State and local regulations.

3.1.2.7 Disposal of Other Materials

Handle other materials than previously discussed (Construction and Demolition, vegetative waste, etc.) as directed by the Contracting Officer.

3.1.3 Preservation and Recovery of Historic, Archeological, and Cultural Resources

3.1.3.1 Applicable Law

A number of Federal laws require protection of cultural resources. Two laws, in particular, can be potentially involved with construction activities: (1) the National Historic Preservation Act, as amended; and, (2) the Abandoned Shipwreck Act.

3.1.3.2 Known Resources

If known historic, archeological, and cultural resources within the Contractor's work area(s) are present, it will be designated as an avoidance area on the contract drawings or other documents. If so designated, install protection for these resources and preserve during the Contract's duration. Do not dredge, spud, or anchor within the designated avoidance area. Do not distribute maps or other information on these resource locations except for distribution among the Contractor's staff with a need to know technical responsibility for protecting the resources.

3.1.3.3 Inadvertent Discoveries

If, during construction activities, items that may have historic or archaeological origin are observed, such observations are to be reported immediately to the Contracting Officer so that the USACE Project Archaeologist may be notified. Cease all activities adjacent to the discovery that may result in the destruction of these resources and prevent employees from further removing, or otherwise damaging, such resources. Once reported, USACE Project Archaeologist will initiate coordination with the appropriate federal, tribal and state agencies to determine if archaeological investigation is required. Additional work in the area of the discovery will be suspended at the site until all federal and state regulations have been successfully completed and the USACE staff members provide further directive. Project activities in the vicinity of the discovery may not resume until the Contracting Officer approves work to proceed.

The possibility of encountering submerged cultural resources is inherent in dredging and snagging operations. Such findings could include shipwrecks, shipwreck debris fields (such as steam engine parts, or wood planks and beams), anchors, ballast rock, concreted iron objects, concentrations of coal, prehistoric watercraft (such as log dugouts), and other structural features intact or displaced. The materials may be deeply buried in sediment, resting in shallow sediments or above them, or protruding into water. Suspected cultural materials inadvertently gathered from a water-saturated context should be kept moist by re-immersion, spraying, or some other expedient means of wetting until the Contracting Officer provide further directives in coordination with the USACE Project Archaeologist. No interviews or other contact with the media or federal, tribal or state agencies are to occur without clear authorization from the Contracting Officer.

In the unlikely event that unmarked human remains are identified, they will be treated in accordance with Section 872.05 Florida Statutes. Immediately cease all work and ground disturbing activities within a 50-meter diameter of the unmarked human remains and notify the Contracting Officer or the USACE representative is to be immediately notified. The USACE Project Archaeologist will notify the local medical examiner, and the appropriate State Historic Preservation Office (SHPO) and Tribal Historic Preservation Office(s) (THPO). Based on the circumstances of the discovery, equity to all parties, and considerations of the public interest, the Contracting Officer may modify, suspend, or terminate work activities. Such activity is not to resume without written authorization from the State Archeologist and from the Contracting Officer.

3.1.3.4 Claims for Downtime due to Inadvertent Discoveries

Upon discovery and subsequent reporting of a possible inadvertent discovery of cultural resources, continue work well away from, or otherwise protectively avoiding, the area of interest, or in some other manner that strives to continue productive activities in keeping with the Contract. If an inadvertent discovery be of the nature that substantial impact(s) to the work schedule are evident, such delays are to be coordinated with the Contracting Officer. Contract adjustments resulting from compliance with this paragraph are to be determined in accordance with Clause DIFFERING SITE CONDITIONS of Section 00700 CONTRACT CLAUSES in Volume 1.

3.1.4 Protection of Water Resources

Keep construction activities under surveillance, management, and control to avoid pollution of surface, ground waters, and wetlands. Implement special management techniques as set out below to control water pollution by the listed construction activities which are included in this Contract. Use construction methods that protect wetland and surface water areas from damage due to mechanical grading, erosion, sedimentation and turbid discharges. Storageing or stockpiling of equipment, tools, or materials within wetlands or along the shoreline within the littoral zone is not allowed unless specifically authorized.

3.1.4.1 Monitoring of Water Areas

Monitor all water areas affected by construction activities.

3.1.4.2 Turbidity

Conduct operations in a manner to minimize turbidity. Refer to Section 01 57 25 TURBIDITY AND DISPOSAL MONITORING for further instructions.

3.1.4.3 Oil, Fuel, and Hazardous Substance Spill Prevention and Mitigation

Prevent oil, fuel, or other hazardous substances from entering the air, ground, drainage, local bodies of water, or wetlands. This is to be accomplished by design and procedural controls. In the event that a spill occurs despite the design and procedural controls, the following is to occur:

(1) Take immediate action to contain and cleanup any spill of oil, fuel or other hazardous substance.

(2) Immediately report spills to the Contracting Officer, the National Response Center (NRC) at 1-800-424-8802. For spills into wetlands or waterways, regardless of quantity, and terrestrial spills over 25 gallons, report to the Florida Department of Environmental Protection's (FDEP) State Warning Point at 1-800-320-0519 within 24 hours. This immediate response must include the suspension of any pumping/dewatering activities that could transport the spilt material to another location. Pumping must be suspended until the spill site is cleaned up and has no visible chemical sheen on the water surface. The Contractor is fully responsible, at no addition cost to the government for the remediation the spill cleanup to include the spill site and any location the spilt material was transferred to.

(3) Spill contingency planning is to be strictly in accordance with the criteria of 40 CFR, Part 109.

(4) To control the spread of any potential spill, absorbent materials are to be readily available and capable of absorbing the contents of the single largest tank.

(5) To control the spread of any potential spill, provide a written certification of commitment of manpower, equipment, and materials required to expeditiously cleanup and dispose of spill materials.

a. Spill Preventive Systems: System design and installation requirements have been discussed at the beginning of this Section. Temporary or portable tanks will conform to applicable Federal, State and local codes and requirements and is not to be placed where they may be affected by storm, flooding, or washout. Diversionary structures for spills are to be put in place in advance where practical. Fuel storage locations to include pumps and generators will be placed and have features such as earthen berms, to prevent any fuel spills to surface water or wetlands. Both spill preventive systems and any deviations from associated requirements will be approved by the Contracting Officer prior to implementation.

b. Liabilities: Liable in the amounts established in 40 CFR, Part 113 when it can be shown that oil was discharged as a result of willful negligence or willful misconduct. The penalty for failure to report the discharge of oil will be in accordance with the provision of 33 CFR, Part 153.

3.1.5 Protection of Fish and Wildlife Resources

Keep construction activities under surveillance, management, and control to minimize interference with, disturbance to, and damage of fish and wildlife. List species that require specific attention along with measures for their protection in the Environmental Protection Plan prior to the beginning of construction operation.

In the event that a threatened, endangered, or Marine Mammal Protection Act (MMPA) species is harmed as a result of construction activities, cease all work and notify the Contracting Officer. The order of contact within the Corps of Engineers will be as follows:

Order of Contact of Corps Personnel

Title	Telephone Number
Corps, Inspector	Onsite/After hours to be provided
Corps, Project Engineer	Onsite/After hours to be provided
Contracting Officer Representative	To be provided
Administrative Contracting Officer	To be provided
Area Engineer for Area Office	To be provided

Information provided above will be updated during the pre-construction meeting.

3.1.5.1 Listed Species Protection

Endangered and Threatened Species and MMPA species that may be in the project area

Common Name	Scientific Name	ESA Status
Florida manatee	Trichechus manatus latirostris	Threatened
North Atlantic Right whale	Eubalaena glacilis	Endangered
Sperm whale	Physeter marcocephalus	Endangered
Humpback whale	Megaptera novaeangliae	MMPA protected
Common Bottlenose Dolphin	Tursiose truncatui	Marine Mammal Protection Act Protected
Shortnose sturgeon	Acipenser brevirostrum	Endangered
Atlantic sturgeon	Acipenser oxyrhynchus oxyrhynchus	Endangered
Green sea turtle	Chelonia mydas	Threatened
Loggerhead sea turtle	Caretta caretta	Threatened
Kemp's ridley sea turtle	Lepidochelys kempii	Endangered
Leatherback sea turtle	Dermochelys coriacea	Endangered
Hawksbill sea turtle	Eretmochelys imbricata	Endangered
Giant Manta Ray	Manta birostris	Threatened
Nassau Grouper	Epinephelus striatus	Threatened
Scalloped hammerhead	Sphyrna lewini	Threatened
Oceanic Whitetip	Carcharhinus longimanus	Threatened
Piping plover	Charadrius melodus	Threatened
Red knot	Calidris canutus rufa	Threatened
Bald eagle	Haliaeetus leucocephalus	Bald and Golden Eagle Act protected
Southeastern Beach Mouse	Peromyscus polionotus niveiventris	Threatened

Instruct personnel associated with the project of the potential presence of listed species as noted in the Table 3.1.5.1 in the area and the need to avoid collisions with and/or harming these animals. All construction personnel are to be advised that there are civil and criminal penalties for harming, harassing, or killing, or attempting to harm, harass or kill or listed species as noted in the Table 3.1.5.1 in the area which are protected under the Marine Turtle Protection Act, Marine Mammal Protection Act of 1972, the Endangered Species Act of 1973, and/or the Florida Manatee Sanctuary Act. The Contractor will be held responsible for any listed species as noted in the Table 3.1.5.1 in the area harmed, harassed, or killed as a result of construction activities.

a. Siltation Barriers: Made of material in which manatees cannot become entangled, properly secured, and regularly monitored to avoid manatee entrapment. Barriers must not block manatee entry to or exit from essential habitat.

b. Special Operating Conditions:

(1) All vessels associated with the project will operate at no wake/idle speeds at all times while in waters where the draft of the vessel provides less than a 4-foot clearance from the bottom, and vessels will follow routes of deep water whenever possible. Boats used to transport personnel must be shallow-draft vessels, preferably of the light-displacement category, where navigational safety permits. Mooring bumpers must be placed on all barges, tugs, and similar large vessels wherever and whenever there is a potential for manatees to be crushed between two moored vessels. The bumpers are to provide a minimum stand-off distance of four feet under maximum designed compression.

(2) Manatee or Marine Turtle Sighting: If a manatee(s) or marine turtle(s) are sighted within 100 yards of the project area, all appropriate precautions will be implemented by the Contractor to ensure protection of the manatee or marine turtle. These precautions will include the operation of all moving equipment, including vessels, no closer than 50 feet of a manatee or marine turtle. If a manatee or marine turtle are closer than 50 feet to moving equipment or the project area, the equipment will be shut down and all construction activities are to cease within the waterway to ensure protection of the manatee or marine turtle. For unanchored vessels, operators are to disengage the propeller and drift out of the potential impact zone. If drifting would jeopardize the safety of the vessel then idle speed may be used to leave the potential impact zone. Construction activities will not resume until animal(s) has moved beyond the 50-foot radius of the project operation, or until 30 minutes elapses if the animals(s) has not reappeared within 50 feet of the operation. Animals will not be herded away or harassed into leaving. If construction activity will cease, notify the Contracting Officer.

(3) Report interactions with threatened or endangered (T&E) species to the Contracting Officer. Cease dredging operations each time an T&E species is potentially taken until the Contracting Officer provide notification to resume dredging. Promptly following the potential take of an T&E species, the Government will perform a risk assessment to include a review of all potential circumstances which contributed to the take, a review of the DQM data, and a physical inspection of the dredge and its operating procedures in accordance with the hopper dredge inspection checklist, project specifications and applicable biological opinions and permits. The Government, in coordination with the Contractor, will develop a risk management plan containing specific actions to minimize risk of additional potential take. The Government may terminate dredging if it is determined that take numbers are at or near the incidental take statement limits or additional take jeopardizes other Government approved regional dredging projects authorized by the South Atlantic/Gulf of America Regional Biological Opinion and the risk of additional take by the project is determined to be high.

(4) If a sea turtle or sturgeon is taken by hopper dredge (dead or alive), a submittal of incident information is to be made into the

Operations and Dredging Endangered Species System (ODESS).

c. Marine Animal Monitoring (Clamshell/Mechanical Only): During clamshell dredging operations, a qualified dedicated observer is to monitor for the presence of manatees and marine turtles. The dedicated observer must be in a location with direct line of sight for all in-water construction activity (i.e. direct line of sight of clamshell bucket or excavator arm entering water). The dedicated observer is to have experience in manatee and marine turtle observation and be equipped with polarized sunglasses to aid in observing. Nighttime lighting of waters within and adjacent to the work area is to be illuminated, using shielded or low-pressure sodium-type lights, to a degree that allows the dedicated observer to sight any manatee or marine turtle on the surface within 200 feet of the operation. The dredge operator must gravity-release the clamshell bucket only at the water surface, and only after confirmation that there are no manatees or marine turtles within the safety distance identified in the standard construction conditions.

d. Manatee Signs: Post temporary signs concerning manatees prior to and during all in-water project activities at sufficient locations to be regularly and easily viewed by all personnel engaged in water-related activities. Remove all signs upon completion of the project. Two temporary signs that have already been approved for this use by the FF&WCC must be used at each location. One sign which reads CAUTION BOATERS - WATCH FOR MANATEES must be posted. A second sign measuring at least 8 ½ inch by 11 inch must explain the requirements for Idle Speed/No Wake and the shutdown of in-water operations. These signs can be viewed at MyFWC.com/manatee.

3.1.5.2 Protected Listed Species Sighting Reports

Report any take concerning any Listed Species in the Table 3.1.5.1; or sighting of any injured or incapacitated of Listed Species to the Contracting Officer by notifying the personnel indicated in the table ORDER OF CONTACT OF CORPS PERSONNEL above.

Provide a copy of the incidental take report in PDF format with photographs to the Contracting Officer, Project Biologist, and Environmental Compliance POC within 24 hours of the incident. Report any collision with and/or injury to a manatee to the Florida Fish and Wildlife Conservation Commission Manatee Hotline 1-888-404-FWCC (3922) and imperiledspecies@myFWC.com, as well as the U.S. Fish and Wildlife Service, Vero Beach Field Office 772-562-3909 for South Florida. Report any collision with (and/or injury to) a marine turtle to the Sea Turtle Stranding and Salvage Network (STSSN) at SeaTurtleStranding@myfwc.com. Report any take of Smalltooth sawfish to l-844-SAW-FISH or email Sawfish@MyFWC.com. Report any take of a Giant manta ray to manta.ray@noaa.gov.

Care must be taken in handling injured marine turtles or eggs to ensure effective treatment or disposition, and in handling dead specimens to preserve biological materials in the best possible state for later analysis. If a marine turtle nest is excavated during construction activities, but not as part of the authorized nest relocation process outlined in the following sections, the permitted person responsible for egg relocation for the project is to be notified immediately so the eggs can be moved to a suitable relocation site.

3.1.5.3 Protected Species Summary Report

If a hopper dredge is used, submit all details regarding potential takes in ODESS and the Quality Control Report. Any incidents, including sightings, collisions with, injuries, or killing of manatees, sea turtles (not associated with a take via the dredge) or whales occurring during the contract period are to be recorded in a log that will be turned into the Contracting Officer at the end of the project. Maintain paper copies of all forms entered into the ODESS until the project is deemed complete by the Contracting Officer. An electronic copy of all incident reports/photos associated with take via the dredge is to be included in the summary report. Weekly Protected Species Summary Report is to be submitted weekly in electronic spreadsheet format to the Contracting Officer and the Project Biologist. An example spreadsheet is available upon request.

Maintain a log detailing all incidents, including sightings, collisions with, injuries, or killing of Listed Species occurring during the contract period. The data is to be recorded on forms provided by the Contracting Officer (sample forms are on the web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below). Within 30 calendar days following project completion, a Protected Species Summary Report summarizing the above incidents and sightings is to be submitted to the Contracting Officer and cc the Environmental Compliance POC.

If a hopper dredge is not used, maintain a log detailing all incidents, including sightings, collisions with, injuries, or killing of manatees, sea turtles or whales occurring during the contract period. The data shall be recorded on forms provided by the Contracting Officer (sample forms are on the web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below). All data in original form shall be forwarded directly to Chief, Environmental Branch, P.O. Box 4970, Jacksonville, Florida, 32232-0019, within 10 days of collection and copies of the data shall be supplied to the Contracting Officer. Within 30 calendar days following project completion, a Protected Species Summary Report summarizing the above incidents and sightings shall be submitted to the following:

Chief, Environmental Branch
U.S. Army Corps of Engineers (CESAJ-PD-E)
P.O. Box 4970
Jacksonville, Florida 32232-0019

Area Engineer for Area Office indicated
on Standard Form 1442

U.S. Fish and Wildlife Service
1339 20th Street
Vero Beach, Florida 32960-3559

National Marine Fisheries Service
Protected Species Management Branch
263 13th Avenue South
St. Petersburg, Florida 33701

Florida Fish and Wildlife Conservation Commission
Imperiled Species Management Section
620 South Meridian Street, Mail Stop 6A
Tallahassee, Florida 32399-1600

3.1.5.4 Endangered Species Observers (Hopper Dredge Only)

During dredging operations, observers approved by the National Oceanic and Atmospheric Administration - Fisheries (NOAA-Fisheries/NMFS) must be aboard to monitor for the presence of the listed species. Observer coverage must be 100 percent (24hr/day) and is to be conducted year round. During transit to and from the disposal area, the observer is to monitor from the bridge during daylight hours for the presence of T&E species, especially the North Atlantic right whale, during the period November 1 through April 30. The ESO(s) are responsible for personally monitoring, handling, and reporting all captured listed species at all times when the hopper dredge is operating. The ESO on-duty will stand watch to detect listed species in the area and to alert the captain of their presence to avoid vessel collision whenever the vessel is moving. The on-duty ESO will only be responsible for standing watch and not performing other tasks such as inspecting or handling captures when the vessel is in motion.During dredging operations, while dragheads are submerged, the observer is to continuously monitor the inflow and/or overflow screening for turtles and/or turtle parts and sturgeon and/or sturgeon parts. Upon completion of each load cycle, dragheads should be monitored as the draghead is lifted from the sea surface and is placed on the saddle in order to assure that listed species that may be impinged within draghead are not lost and un-accounted for. Observers are to physically inspect dragheads and inflow and overflow screening/boxes for threatened and endangered species take. Hopper dredge operators will not open the hydraulic doors on the inflow boxes prior to inspection by the ESO for evidence of listed species take. The screened area must be accessible to the ESO to ensure 100% observer coverage. The ESO must inspect the contents of all inflow screening boxes after every load, including opening the box (where applicable and safely accessible) and looking inside at all contents for evidence of listed species entrainment. If the contents are not clearly visible and identifiable from a location outside of the box, then in limited instances, the ESO may be required to enter the inflow box to identify contents. If the inflow box cannot be observed due to clogging, the box contents cannot be dumped or flushed unless overflow screening that captures contents for observation by the ESO is operational and monitored for evidence of take. ESOs are to visually monitor box contents as they are dumped or flushed into the hopper.

a. Monitoring Reports: The results of the monitoring are to be recorded on the appropriate observation sheets and entered into the ODESS system (See Appendix A attached after the end of this Section) and Quality Control Daily Report. There is a form for each load and a form for an incident involving a listed species, specific to the species (turtles, sturgeon, etc.). Complete load forms regardless of whether any incidents with sturgeon (Gulf, Shortnose or Atlantic), whales, or sea turtles occur. In the event of any sea turtle or sturgeon (Gulf, Atlantic or Shortnose) is potentially taken by the dredge, appropriate incident reporting forms are to be completed and entered into the ODESS system. Additionally, photograph all specimens with the digital camera built into the ODESS tablet. Upload these photographs to the respective incident reports on the ODESS within 4-hours from observing the incident. Do not commence dredging of subsequent loads until all appropriate reports are completed from the previous dredging load to ensure completeness and thoroughness of documentation associated with the incidental take exports. Submit reports to the Contracting Officer within 4-hours of the take. Observer forms can be accessed on the web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below.

b. Digital Photographs: Use a digital camera embedded in the ODESS tablet, with an image resolution capability of at least 300 dpi, in order to photographically report all incidental takes, without regard to species, during dredging operations. Upload photos taken associated with the take to the ODESS with the incident information.

c. A copy of all incident forms, load forms, and photographs from incidents with listed species is to be included in the Protected Species Summary Report.

3.1.5.5 Disposition of Endangered or Threatened Species Carcasses

a. Listed Species Taken by Hopper Dredge:

(1) Dead ESA-listed species including Turtles or sturgeon (regardless of species): Upon removal of carcass and/or parts from the draghead or screening, observers are to take photographs as to sufficiently document major characteristics of the carcass or parts including but not limited to dorsal, ventral, anterior, and posterior views. For all photographs taken, a backdrop is to be prepared to document the dredge name, observer company name, contract title, time, date, species, load number, location of dredging, and specific location taken (draghead, screening, etc.). Carcass/parts are to be scanned for flipper and Passive Integrated Transponder (PIT) tags. Any identified tags are to be recorded on the ODESS Turtle Incident Form or ODESS Sturgeon Incident Form (as appropriate) that is included in the ODESS Program Forms available through the ODESS tablet (or on the ODESS website if the tablet is not available). Also, fill out the ODESS Dredge Load Form and take photos for submission. Animal parts which cannot be positively identified to species, on board the dredge or barge(s) are to be preserved by the observer(s) for later identification. A tissue sample must be collected from any lethally taken ESA species submitted under the process stated in the Protocol for Collecting Tissue Samples from Turtles for Genetic Analysis found in the CONSTRUCTION FORMS AND DETAILS below. All genetic samples collected are to be submitted to NMFS within 30 days of collection, and verification of submittal to NMFS is to be provided to the Contracting Officer and USACE Project Biologist. After all data collection is complete, the carcass/parts should be marked (spray paint works well), weighted down and disposed of, or provided to a state agency staff member, in accordance with the direction of the Contracting Officer.

(2) Live ESA-listed species including Turtles: Observer(s) is to measure, weigh, scan for PIT tags, tag (Iconnel flipper and PIT tags (if PIT tag not located during scan, and only if observer is qualified to tag using PIT tags)), and photograph any live listed species incidentally taken by the dredge. Observer(s) (or their authorized representative) is to coordinate with the Contracting Officer and Environmental branch staff prior to transport. Handling of any marine turtles captured during hopper dredging is to be conducted only by persons with prior experience and training in these activities, such as a NMFS-approved marine turtle observer, or by persons who have submitted documentation to the Contracting Officer of meeting the FWC Marine Turtle Conservation Guidelines specific to stranding activities. The Contracting

Officer or their designee who transport live or dead marine turtles or marine turtle parts into, out of, or within, the state of Florida will notify FWC in writing specifying the number, species of turtle, type of specimen, and the destination after transport is complete. Before transport, the Contracting Officer or their designee must coordinate with FWC at SeaTurtleStranding@myfwc.com to determine the appropriate facility to receive live marine turtles for rehabilitation. The Contracting Officer or their designee will abide by the State of Florida's FWC Marine Turtle Conservation Guidelines (https://myfwc.com/media/3133/fwc-mtconservationhandbook.pdf) specific to transport of live stranded turtles.

b. Turtles Taken by Relocation Trawler:

(1) Live Turtles: At least one (1) crewmember, who possesses the required permits for handling endangered species, experienced in sea turtle capture or is a NMFS-approved observer, is to be on board the trawler during the trawl and act as the sea turtle trawling and relocation supervisor. Only a NMFS-approved observer or observer candidate in training under the direct supervision of a NMFS-approved observer is to conduct the sampling operations of tagging, measuring, weighing and collecting tissue samples from turtles. Each turtle that is captured is to be identified to species and age class (juvenile, sub-adult, adult), digitally photographed, scanned for Passive Integrated Transponder (PIT) tags, measured, tagged (Inconel flipper tag and PIT tag, if one not located by scan, and only if observer is qualified to tag using PIT tags), and released and the data recorded on the Sea Turtle Tagging and Relocation Report found in the CONSTRUCTION FORMS AND DETAILS below. External tags are to also be noted and data recorded into the observer's log. Turtles are to be released at locations at least 3 nautical miles (nmi) away from the project area, at time intervals designated by the Sea Turtle Trawling and Relocation Supervisor who is proficient in the viable handling and relocating of sea turtles.

Weight/Size Measurements and Tissue Collection. Turtles must be measured, tagged, and weighed when safely possible, prior to release. Turtle measurements is to be recorded and must include, at a minimum, weight when possible, straight-line length, straight-line width, body depth, and tail length. Turtles are to be tagged with NMFS No. 681 Inconel tags in each of the front flippers and PIT tags according to NMFS protocol (found at https://www.sefsc.noaa.gov/turtles/TM_579_SEFSC_STRTM.pdf and on the ODESS website under the Information tab). Aseptic conditions are maintained for tags and tag attachment. A tissue sample is to be collected from any trawl captured turtle and submitted under the process stated in the Protocol for Collecting Tissue Samples from Turtles for Genetic Analysis found in the CONSTRUCTION FORMS AND DETAILS below. All genetic samples collected are to be submitted to NMFS within 30 days of collection, and verification of submittal to NMFS is to be provided to the Contracting Officer. Obtain permits related to trawling from the appropriate State agencies.

(2) Dead Turtles: Any turtle found dead in the nets during relocation trawling efforts is to be digitally photographed, measured, weighed (if possible), scanned for tags. The carcass should then

be placed on ice and kept cold (but not frozen) as quickly as possible and the Contracting Officer is to be contacted immediately. The Contracting Officer will contact the Florida State Sea Turtle Salvage and Stranding network (SeaTurtleStranding@myFWC.com) to arrange for the carcass to be transferred for necropsy. The turtle is to be kept on ice (but not frozen) until the appropriate state agency can be contacted. If the state agency declines to recover the carcass for necropsy, the carcass is to be marked (spray paint works well), weighted down and disposed of in accordance with the direction of the Contracting Officer.

Weight/Size Measurements and Tissue Collection. Turtles are to be measured, and weighed. Measurements are to be recorded and must include, at a minimum, weight when possible, straight-line length, straight-line width, body depth, and tail length. The Carcass is to also be scanned for flipper and Passive Integrated Transponder (PIT) tags. Any identified tags are to be recorded on the Sea Turtle Incidental Take Form that is included in the Endangered Species Observer Program Forms located on the web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below. A tissue sample is to be collected from any lethally taken sea turtle and submitted under the process stated in the Protocol for Collecting Tissue Samples from Turtles for Genetic Analysis found in the CONSTRUCTION FORMS AND DETAILS below. All genetic samples collected are to be submitted to NMFS within 30 days of collection, and verification of submittal to NMFS is to be provided to the Contracting Officer. Obtain permits related to trawling from the appropriate State agencies.

3.1.5.6 Hopper Dredge Equipment

Hopper dredge dragheads are to be equipped with rigid sea turtle deflectors which are rigidly attached. Deflectors are to be solid with no openings in the face. Alternative designs will be considered provided sufficient information is included indicating a particular modification is effective in minimizing potential turtle takes. The Government technical staff will coordinate with NMFS on the effectiveness of this alternate design. A recommendation from the Government technical staff and NMFS will be provided to the Contracting Officer for final approval or disapproval of the alternate design. Do not presume that a decision on an alternative design will be provided during the contracting period. The Contractor's unit price is to be based on the original, solid faced deflector design. No adjustment in unit price will be made for the approval or denial of an alternate deflector design. No dredging is to be performed by a hopper dredge without an installed turtle deflector device approved by the Contracting Officer. Sample Turtle Deflector Design Details are on the first web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below. For each dredge operating on the project, dredging operations must not begin until the USACE Sea Turtle Deflector Checklist for Hopper Dredges for USACE and USACE/Army-permitted projects has been completed by a USACE-approved inspector. The completed checklist for each dredge is to be submitted by the USACE-approved inspector to the Contracting Officer and forwarded to the Project Biologist. The checklist can be downloaded from the ODESS website - https://odess.usace.army.mil

If a hopper dredge is used for this work, submit detail drawings showing the proposed Turtle Deflector Device and its attachment to the equipment no later than 30 calendar days after notice to proceed. Include the approach angle for any depths to be dredged during this Contract on the submitted drawing. A copy of the approved drawings and calculations are to be available on the vessel during the life of this Contract. Do not commence any dredging work until approval of the installed turtle deflector device.

a. Deflector Design:

(1) The leading vee-shaped portion of the deflector is to have an included angle of less than 90 degrees. Internal reinforcement is to be installed in the deflector to prevent structural failure of the device. The leading edge of the deflector is to be designed to have a plowing effect of at least 6 inch depth when the drag head is being operated. Appropriate instrumentation or indicator is to be used and kept in proper calibration to insure the critical approach angle. (Information Only Note: The design approach angle or the angle of lower drag head pipe relative to the average sediment plane is very important to the proper operation of a deflector. If the lower drag head pipe angle in actual dredging conditions varies tremendously from the design angle of approach used in the development of the deflector, the 6 inch plowing effect does not occur. Therefore, every effort should be made to ensure this design approach angle is maintained with the lower drag pipe.)

(2) If adjustable depth deflectors are installed, they must be rigidly attached to the drag head using either a hinged aft attachment point or an aft trunnion attachment point in association with an adjustable pin front attachment point or cable front attachment point with a stop set to obtain the 6 inch plowing effect. This arrangement allows fine-tuning the 6 inch plowing effect for varying depths. After the deflector is properly adjusted there is to be no openings between the deflector and the drag head that are more than 4 inches by 4 inches.

(3) Draghead Screening: If the Contractor has the draghead(s) screened, the screening for the draghead production opening must be 8 inches by 8 inches or larger Pumps employed must be sufficient to operate efficiently with this screening as a minimum.

b. InFlow/Overflow Basket Design:

(1) Install baskets or screening over the hopper inflow/overflow(s) with no greater than 4 inch by 4 inch openings. The method selected is to depend on the construction of the dredge used and is to be approved by the Contracting Officer prior to commencement of dredging. The screening is to provide 100 percent coverage of the hopper inflow and overflow(s). The screens and/or baskets are to remain in place throughout the performance of the work. 100 percent inflow and overflow screening is to be maintained throughout the project. If 100 percent inflow and overflow screening cannot be maintained due to clogging of the baskets by dredged material, or other material including reeds, shell, etc, coordinate with the Contracting Officer to determine an appropriate path forward.

(2) Install and maintain floodlights suitable for illumination of the baskets or screening to allow the observer to safely monitor the hopper basket(s) during non-daylight hours or other periods of

poor visibility. Safe access is to be provided to the inflow baskets or screens to allow the observer to inspect for turtles, turtle parts or damage.

c. Hopper Dredge Operation:

(1) Operate the hopper dredge to minimize the possibility of taking sea turtles or sturgeon and to comply with the requirements stated in the Incidental Take Statement provided by the NMFS in the appropriate Biological Opinion.

(2) The turtle deflector device and inflow and overflow screens are to be maintained in operational condition for the entire dredging operation.

When initiating dredging, suction through the drag heads is to be allowed just long enough to prime the pumps, then the drag heads are to be placed firmly on the bottom. When lifting the drag heads from the bottom, suction through the drag heads must be allowed just long enough to clear the lines, and then must cease. Pumping water through the drag heads is to cease while maneuvering or during travel to/from the disposal area. (Information Only Note: Optimal suction pipe densities and velocities occur when the deflector is operated properly. If the required dredging section includes compacted fine sands or stiff clays, a properly configured arrangement of teeth may enhance dredge efficiency which reduces total dredging hours and turtle takes. The operation of a drag head with teeth is to be monitored for each dredged section to ensure that excessive material is not forced into the suction line.

(3) When excess high-density material enters the suction line, suction velocities drop to extremely low levels causing conditions for plugging of the suction pipe. Dredge operators should configure and operate their equipment to eliminate all low level suction velocities. Pipe plugging cannot be corrected by raising the drag head off the bottom. Arrangements of teeth and/or the reconfiguration of teeth should be made during the dredging process to optimize the suction velocities.)

(4) Raising the drag head off the bottom to increase suction velocities is not acceptable. The primary adjustment for providing additional mixing water to the suction line should be through water ports. To ensure that suction velocities do not drop below appropriate levels, monitor production meters throughout the job and adjust primarily the number and opening sizes of water ports. Water port openings on top of the drag head or on raised stand pipes above the drag head is to be screened before they are utilized on the dredging project. If a dredge section includes sandy shoals on one end of a tract line and mud sediments on the other end of the tract line, adjust the equipment to eliminate drag head pick-ups to clear the suction line.

(5) Near the completion of each payment section, perform sufficient surveys to accurately depict those portions of the acceptance section requiring cleanup. Keep the drag head buried a minimum of 6 inches in the sediment at all times. Although the over depth prism is not the required dredging prism, achieve the required prism by removing the material from the allowable over depth prism.

(6) During turning operations the pumps are to either be shut off or reduced in speed to the point where no suction velocity or vacuum exists.

(7) These operational procedures are intended to stress the importance of balancing the suction pipe densities and velocities in order to keep from taking sea turtles. Develop a written operational plan to minimize turtle takes and submit it as part of the Environmental Protection Plan.

(8) Comply with all requirements of this specification and the accepted Environmental Protection Plan. The contents of this specification and the Environmental Protection Plan is to be shared with all applicable crew members of the hopper dredge.

(9) To prevent impingement or entrainment of listed species within the water column, dredging pumps will be disengaged by the operator when the dragheads are not actively dredging and therefore working to keep the draghead firmly on the bottom. Pumps will be disengaged when lowering dragheads to the bottom to start dredging, turning, or lifting dragheads off the bottom at the completion of dredging. Hopper dredges may utilize a bypass or other system that would allow pumps to remain engaged, but result in no suction passing through the draghead. Pumping water through the dragheads is not allowed while maneuvering or during travel to/from the disposal or pumpout area. The dredge operator will ensure the draghead is embedded in sediment when pumps are operational, to the maximum extent practicable.

(10) All waterport or other openings on the hopper dredge are required to be screened to prevent listed species from entering the dredge.

d. Hopper Dredge Lighting Requirements: From March 1st through November ~~11~~30th, sea turtle nesting and emergence season, all lighting aboard hopper dredges and hopper dredge pumpout barges operating within three nautical miles of sea turtle nesting beaches is to be limited to the minimal lighting necessary to comply with U.S. Coast Guard and Occupational Safety and Health Administration (OSHA) requirements. All non-essential lighting on the dredge and pumpout barge is to be minimized through reduction, shielding, lowering and appropriate placement of lights to reduce potential disorientation effects on female sea turtles approaching the nesting beaches and sea turtle hatchlings heading seaward.

3.1.5.7 Sea Turtle Trawling and Relocation (For Hopper Dredges Only)

a. Sea Turtle Trawling and Relocation: Initiation of sea turtle relocation via trawling will be determined by the Contracting Officer following the take of two sea turtles, without regard to species, and continue until as directed by the Contracting Officer. The results of each trawl is to be recorded on Sea Turtle Trawling Report available from the ODESS website (https://odess.usace.army.mil/). An interim trawling report is to be submitted within 24 hours after the third turtle take by the dredge. A final report is to be prepared after the completion of all trawling efforts. Both reports are to be submitted in hard copy and electronic format to the Contracting Officer, and must summarize the results of the trawling (with all forms and including total trawling times, number of trawls and number of captures). Any turtles captured during the trawling effort are to be measured and tagged in accordance with standard biological sampling procedures with sampling data recorded on Sea Turtle Tagging and Relocation Report available from the ODESS website (https://odess.usace.army.mil/). Any captured sea turtles are to be relocated from the work area no less than 3 miles away from the location recorded on the Sea Turtle Tagging and Relocation Report form.

b. Sea Turtle Trawling Procedures: A NMFS approved sea turtle trawling and relocation supervisor is to provide sea turtle trawl observers and nets to capture and relocate sea turtles and is to conduct any initiated sea turtle trawling. Turtles are to be captured with trawl nets to relocate them away from the project area to reduce the potential of sea turtle take. Methods and equipment is to be standardized including nets, trawling direction to tide, length of station, length of tow, and number of tows per station. Data on each tow is to be recorded using Sea Turtle Trawling Report available from the ODESS website (https://odess.usace.army.mil/). The trawler is to be equipped with two 60-foot nets constructed from 8-inch mesh (stretch) fitted with mud rollers and flats as specified in Turtle Trawl Nets Specifications on the first web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below. Paired net tows are to be made for 10 to 12 hours per day or night. Trawling is to be conducted with the tidal flow using repetitive 15-30 minute (total time) tows in the channel or borrow area. Tows are to be made as close to the operating dredge as the trawler captain determines is safe. Positions at the beginning and end of each tow are to be determined from GPS Positioning equipment. Tow speed must be recorded at the approximate midpoint of each tow. Refer to EM 1110-1-1003, paragraph 5.3 and Table 5-1, for acceptable GPS criteria.

c. Water Quality and Physical Measurements: Water temperature measurements are to be taken at the water surface each day using a laboratory thermometer or a thermometer built into the trawling vessel. Weather conditions are to be recorded from visual observations and instruments on the trawler. Weather conditions, air temperature, wind velocity and direction, sea state-wave height, and precipitation is to be recorded on the Sea Turtle Trawling Report on the second web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below. High and low tides are to be recorded.

d. Initiation of Trawling: Trawling will be initiated at the discretion of the Contracting Officer. Initiate trawling and relocation activity in the dredging area as quickly as possible (within 48-72 hours) from the notification of the Contracting Officer. Continue trawling until suspended by the Contracting Officer.

e. Approved Trawling Supervisor: Trawling is to be conducted under the supervision of a biologist approved by the NMFS. A letter of approval from NMFS is to be provided to the Contracting Officer prior to commencement of trawling. Approved trawl supervisors can be verified through NMFS-NERO (978)-281-9291.

f. Turtle Excluder Devices: Approval for trawling for sea turtles without Turtle Excluder Devices (TEDs) must be obtained from NMFS. Approval for capture and relocation of sea turtles must be obtained from the Florida Fish and Wildlife Conservation Commission (FF&WCC). Approvals must be submitted to the Contracting Officer prior to trawling.

g. Daily Reporting: A daily log is to be kept for each capture trawling operation. The results of each trawl are to be recorded on the Sea Turtle Trawling Report posted on the second web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below. Positions at the beginning and end of each tow are to be determined from GPS positioning equipment. Tow speed is to be recorded at the approximate midpoint of each tow. See EM 1110-1-1003, paragraph 5.3 and Table 5-1, for acceptable GPS criteria. Data to be recorded in the comments field must also include:

h. Weekly Logs/Summary Of FWC Hopper Dredge Reports: The Contractor is to e-mail the Contracting Officer, (MTP@MyFWC.com), and cc the Environmental Compliance POC weekly reports to the Imperiled Species Management section on Friday each week that trawling is conducted in Florida waters. These weekly reports are to include the species and number of turtles captured in Florida waters, general health, and release information. The Contractor is to provide a summary (on FWC provided Excel spreadsheet) of the following: all trawling activity, all turtles captured in Florida waters, all measurements, the latitude and longitude (in decimal degrees) of captures and tow start-stop points, and times for the start-stop points of the tows (including those tows on which no turtles are captured).

i. Comprehensive Trawling Daily Logs: Within 30 calendar days of completion of the project, submit a copy of the logs regarding capture trawling to the Contracting Officer, (MTP@MyFWC.com), and cc the Environmental Compliance POC.

3.1.5.8 Sea Turtle Beach Nest Monitoring

a. Sea Turtle (Work Stoppage) Window and Monitoring: Conduct work on the beach (sand placement, grading, mobilization of equipment) only from November 1st through April 30th. If work on the beach begins during the period from November 1st through January 15th, begin sea turtle monitoring and nest location/relocation 65 days prior to the commencement of work on the beach and continue concurrently with performance of work. If no nest has been located/relocated for the ~~seventh consecutive day on or after November 1st~~ seven consecutive days, the Contracting Officer may cease monitoring activities upon receipt of electronic mail concurrence from the Contracting Officer. If work on the beach begins during the period from January 16th through February 28th, no monitoring or nest location/relocation is required until March 1st. If work on the beach begins on or after March 1st, begin sea turtle monitoring and nest location/relocation on March 1st and continue concurrently with performance of work. Complete all work on the beach, including demobilization of equipment and final dressing, by the end of the day on April 30th.

b. During the period from February 15 through April 30, daytime surveys must be conducted for leatherback sea turtle nests beginning February 15. Nighttime surveys for leatherback sea turtles must begin when the first leatherback crawl is recorded within the project or adjacent beach area through April 30 or until completion of the project (whichever is earliest). Nightly nesting surveys must be conducted from 9 p.m. until 6 a.m. The project area is to be surveyed at 1-hour intervals (since leatherbacks require at least 1.5 hours to complete nesting, this will ensure all nesting leatherbacks are encountered) and eggs must be relocated. A log of the results of turtle egg monitoring and recovery activities must be kept and a copy submitted weekly to the Chief, Environmental Branch, Jacksonville District (sample Marine Turtle Nesting Summary Report form is on the web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below).

c. Daily Visual Inspection: Turtle monitoring activities are to include performance of daily visual inspections of the beach at sunrise by a person permitted by the FF&WCC for handling sea turtle eggs. Any nests discovered are to be excavated and relocated prior to 9:00 a.m. to a nearby self-release beach location where artificial lighting and/or other disturbances is to not interfere with successful incubation, hatching nor hatchling orientation. A Log Of The Results Of Turtle Egg Monitoring And Recovery Activities is to be kept and a copy submitted weekly to the Contracting Officer, Project Biologist and cc the Environmental Compliance POC (sample Marine Turtle Nesting Summary Report form is on the first web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below).

d. Fill Restrictions: Between March 1 to November ~~11~~30, the Contractor is to not advance the beach fill more than 500 feet along the shoreline between dusk and the following day, until the daily nesting survey is completed, and the beach has been cleared for fill advancement. If the 500-foot advancement limitation is not feasible for the project, an alternative distance is to be established during the preconstruction meeting, if a distance can be agreed upon in consultation with the Contracting Officer and FWC. If the work area is extended, nighttime nesting surveys are required, and a Marine Turtle Permit Holder is required to be present on-site to ensure that no nesting and hatching marine turtles are present. If any nesting turtles are sighted on the beach within the immediate construction area, activities are to cease immediately until the turtle has returned to the water and the Marine Turtle Permit Holder responsible for nest monitoring has relocated the nest.

e. Turtle Subcontractor: Provide a FF&WCC permitted subcontractor approved by the Contracting Officer to accomplish the sea turtle monitoring and relocation of this section unless he demonstrates to the satisfaction of the Contracting Officer the capability to accomplish sea turtle monitoring and recovery by obtaining a permit from the FF&WCC to take turtles.

f. Report Submission: Within 30 days of completion of the project, submit the Logs/Summary of Marine Turtle Nesting and Relocation to the Project Biologist, Contracting Officer and cc the Environmental Compliance POC. This report is to include the project location (FDEP-R monuments), dates of construction, descriptions and locations of self-release sites.

3.1.5.9 Beach Placement Restrictions

a. Equipment Lighting During Sea Turtle Nesting Period March 1 to November ~~11~~30: Direct lighting of the beach and near shore waters is to be limited to the immediate construction area and is to comply with safety requirements. Lighting on offshore or onshore equipment is to be minimized through reduction, shielding, lowering, and appropriate placement to avoid excessive illumination of the waters surface and nesting beach while meeting all Coast Guard, EM 385-1-1, and OSHA requirements. In addition, comply with IES RP-12 and IES RP-16. Light intensity of lighting plants should be reduced to the minimum standard required by OSHA for General Construction areas, in order not to misdirect sea turtles. Shields should be affixed to the light housing and be large enough to block light from all lamps from being transmitted outside the construction area. Refer to Beach Lighting Schematic on the first web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below.

b. Beach Maintenance. The Contractor is to inspect all excavations and temporary alteration of beach topography (i.e., depressions, ruts, holes and vehicle tracks) for the presence of protected species (particularly flightless shorebird chicks or marine turtle hatchlings) and to fill or level any deviations from the natural beach profile prior to 9:00 p.m. each day outside the active construction area. The beach surface is to also be inspected subsequent to completion of the project, and all tracks, mounds, ridges or impressions left by construction equipment on the beach is to be smoothed and leveled.

c. All debris, including derelict coastal armoring material, concrete and metal is to be removed from the beach prior to any placement of construction material on the beach to the maximum extent practicable. If debris removal activities will take place during protected species nesting seasons (~~March 1st through November 11th~~February 15th through November 30th), the work is to be conducted during daylight hours only, and is not to commence until completion of daily monitoring surveys. Note: when flightless chicks are present within or adjacent to travel corridors, construction-related vehicles must not be driven through the corridor unless a Bird Monitor is present.

d. Predator-proof trash receptacles is to be installed and maintained at all beach access points used for the project construction to minimize the potential for attracting predators of sea turtles.

e. Equipment Storage and Placement. Staging areas for construction equipment is to be located off the beach, if off-beach staging areas are available, during the sea turtle and shorebird nesting season ~~(April 1st through September 1st)~~(February 15th through November 30th). Nighttime storage of construction equipment not in use is to be off the beach to minimize disturbance to sea turtle nesting and hatching activities. In addition, all construction pipes that are placed on the beach is to be located as far landward as possible without compromising the integrity of the existing or reconstructed dune system. Pipes placed parallel to the dune are to be 5 to 10 feet away from the toe of the dune. Temporary storage of pipes is to be off the beach to the maximum extent possible. If the pipes are on the beach, they must be placed in a manner that will minimize the impact to nesting habitat and will not compromise the integrity of the dune systems. If it will be necessary to extend construction pipes past a known shorebird nesting site or over-wintering area for piping plovers, then whenever possible, those pipes should be placed landward of the site before birds are active in that area. Should pipe or sand need to be placed seaward of a known shorebird nesting site during shorebird nesting season (~~April 1st - September 1st~~(February 15th through August 31st), the FWC is to be consulted prior to any placement.

f. Beach Tilling: All tilling activity is to be completed prior to demobilization. If the project is completed during the sea turtle nesting season (March 1st through November ~~11th~~30th), tilling is not be performed in areas where nests have been left in place or relocated to. If tilling occurs during shorebird nesting season (~~April 1st - September 1st~~February 15th through August 31st), shorebird surveys are required prior to tilling. Avoid tilling in areas where nesting birds are present; no tilling is to occur within 300 feet of any shorebird nest. If flightless shorebird young are observed within the work zone or equipment travel corridor, a Shorebird Monitor is to be present during the operation to ensure that equipment does not operate within 300 feet of the flightless young.

The fill area is to be tilled to a depth of 36 inches. Each pass of the tilling equipment is to overlap to allow thorough and even tilling. A relatively even surface, with no deep ruts or furrows, is to be created during tilling. Chain-linked fencing or other material is to be dragged over those areas as necessary after tilling. Tilling is to occur landward of the wrack line (typical wave uprush) to the seaward edge of the dune. Avoid all vegetated areas 3 square feet or greater with a 3-foot buffer around the vegetated areas. The slope between the mean high water line and the mean low water line is to be maintained in such a manner as to approximate natural slopes. Any vehicles operated on the beach in association with tilling is to operate in accordance with the FWC's Best Management Practices for Operating Vehicles on the Beach (http://myfwc.com/conservation/you-conserve/wildlife/beach-driving/).

g. Beach Driving. All vehicles are to be operated at speeds less than 6 mph, and run at or below the high-tide line. All personnel associated with the project are to be instructed about the potential presence of onsite protected species, and the need to avoid injury and disturbance to these species. In addition, all vehicles operated on the beach are to operate in accordance with the FWC’s Best Management Practices for Operating Vehicles on the Beach (https://myfwc.com/conservation/you-conserve/wildlife/beach-driving/).

3.1.5.10 Escarpments

If any escarpments exceeding 18 inches in height for a distance of at least 100 feet are found in the placement area, immediately notify the Contracting Officer and provide the locations, dimensions (including latitude and longitude), and photographs of the escarpments. The Contracting Officer will inform the Contractor of the appropriate action to be taken.

a. During Placement Escarpment Monitoring and Removal: Perform daily visual surveys prior to 8 AM for escarpments along finished sections of the beach placement area. During the sea turtle nesting season, perform the survey immediately after the sea turtle monitor gives the all clear. If the Contracting Officer directs the Contractor to remove the escarpment, then the slope of the removal is to be no steeper than lv:l0h. Perform the removal with a bulldozer or similar equipment. Perform the removal the same day that the escarpment is identified. Utilize the low tide period to the greatest extent practical to remove the escarpment and achieve the lv:l0h slope. Complete the removal at least one half hour before sunset.

b. Protection of Sea Turtle Nests: Identify any sea turtle nests located in the vicinity of an escarpment removal and is to protect the nests from disturbance or damage during removal operations.

c. Refer to Protection of Shorebird and Shorebird Surveys and Shorebird

Buffer Zones and Travel Corridors below for shorebird nesting monitoring requirements when performing project activities on the beach such as removing escarpments during the shorebird nesting season (~~April 1st - September 1st~~February 15th through August 31st).

d. Escarpment Site Visit Report(s): Within 5 days after each site visit to monitor for or remove escarpments, provide to the Contracting Officer and cc the Environmental Compliance POC a written report of the visit which indicates the nature of the visit, location and dimensions of any escarpments encountered, whether the escarpments were removed, the beginning and ending point locations (lat/longs or other GPS format) for the removal activity, the equipment used for removal, before and after photographs of the removal area, whether any sea turtle nests were present near the escarpments, and the actions taken to protect the nearby nests.

3.1.5.11 Hardbottom Protection

Hardbottom resources are present within and around the work area. Information of the location of these resources are provided in the plans, but the Contractor is responsible for the identification and protection of these resources outside of the pipeline corridors. Mitigation for hardbottom areas located within the designated fill limits has been previously completed. The Contractor is not responsible for performing additional mitigation for fill placed within these specified limits. Install all protection for these resources so designated on the drawings and must be responsible for their preservation during this contract. Refer to Section 35 20 25 BEACH FILL.

Prior to project commencement, scientific divers are to swim the pipeline corridors at all potential hardbottom crossings to help select alignments that would minimize potential impacts to nearshore hardbottom. Hardbottom surveys of the selected pipeline alignments must be conducted prior to pipeline placement and shortly after removal in order to quantify any impacts. The Pre-Construction Pipeline Corridor Hardbottom Survey must be submitted to the Contracting Officer no later than 55 calendar days prior to pipeline placement and the Pre-Construction Pipeline Corridor Hardbottom Survey Report no later than 40 days prior to pipeline placement. The Post-Placement Pre-Pumping Pipeline Survey Results are to be submitted no later than 96 hours prior to the intended or actual start of pumping. The Post-Construction Pipeline Corridor Hardbottom Survey are to be submitted to the Contracting Officer within 30 days after completion of construction. All pipeline surveys must meet the requirements of the July 2021 St. Lucie County South Beach Restoration, Permit No. 0154626-001-JC Biological Monitoring Plan as Appendix B after the end of this section.

Scientific divers are to also assist in the placement of any submerged pipelines to ensure that submerged pipelines are not placed over any hardbottom/reef areas. If pipeline leaks, mis-dumps, or spillage occurs, then scientific divers must be used to verify the condition of nearby hardbottom areas and the location of the pipeline after it is installed.

3.1.5.12 Protection of Shorebird and Shorebird Surveys

For simplicity, the term “shorebird” is used here to refer to all solitary nesting shorebirds and colonial nesting seabirds that nest on Florida beaches. If any project activities as described below are conducted, the following shorebird protection conditions are required during the shorebird breeding cycle, which includes nesting. The following conditions are intended to avoid direct impacts associated with the construction of the project, and may not address all potential take incidental to the operation and use related to this authorization.

Shorebird breeding season dates for this project area are ~~April 1~~February 15th through September 1 (note that while most species have completed the breeding cycle by September 1, flightless young may be present through September and must be protected if present).

Any parts of the project where "project activities" on the beach take place entirely outside the breeding season, do not require shorebird surveys. The term "project activities" includes operation of vehicles on the beach, movement or storage of equipment on the beach, sand placement or sand removal, and other similar activities that may harm or disturb shorebirds. Breeding season surveys are to begin on the first day of the breeding season or 10 days prior to project commencement (including survey activities and other pre-construction presence on the beach), whichever is later.

Bird surveys are to be conducted in all potential beach-nesting bird habitats within the project boundaries that may be impacted by construction or pre-construction activities. One or more shorebird survey routes are to be established by the Contractor to cover project areas that require shorebird surveys. These routes are to be determined in coordination with the Contracting Officer prior to the initiation of construction. Routes must not be modified without prior notification to the Contracting Officer.

During all pre-construction and construction activities associated with the project, the Contractor is to ensure that surveys for detecting breeding activity and the presence of flightless chicks are to be completed on a daily basis by a qualified bird monitor prior to movement of equipment, operation of vehicles, or other activities that could potentially disrupt breeding behavior or cause harm to the birds or their eggs or young. If all project activities are completed and all personnel and equipment have been removed from the beach prior to the end of the breeding season, route surveys must continue to be conducted at least weekly through the end of the breeding season. Surveys are to also determine whether the completed project has attracted birds to the renourished area and appropriate shorebird buffer zones are to be installed and removed as per required in this authorization. If breeding or nesting behavior is confirmed by the presence of a scrape, eggs or young, the Contractor is to establish a 300-foot buffer around the site and notify the Contracting Officer within 24 hours. The posts and materials for the shorebird buffer zones are to be removed once all breeding or nesting behavior has ceased.

The Bird Monitor is to conduct a shorebird education and identification program (and/or provide educational materials) with the on-site staff to ensure protection of precocial (mobile) chicks. All personnel are responsible for watching for shorebirds, nests, eggs and chicks. If the Bird Monitor finds that shorebirds are breeding within the project area, a bulletin board is to be placed and maintained in the construction staging area with the location map of the construction site showing the bird breeding areas and a warning, clearly visible, stating that "NESTING BIRDS ARE PROTECTED BY LAW INCLUDING THE FLORIDA ENDANGERED AND THREATENED SPECIES ACT AND THE STATE and FEDERAL MIGRATORY BIRD ACTS".

Bird survey protocols, including downloadable field data sheets, are available on the FSD website. All breeding activity is to be reported to the FSD website within one week of data collection. If the use of this website is not feasible for data collection, the FWC Regional Biologist is to be contacted for alternative methods of reporting. The Contractor is to ensure that the Bird Monitors use the following survey protocols:

a. Surveys are to be conducted by walking the length of all survey routes and visually surveying for the presence of shorebirds exhibiting breeding behavior, shorebird chicks or shorebird juveniles, as outlined in the FSD Breeding Bird Protocol for Shorebirds and Seabirds. Use of binoculars (minimum 8x40) is required, and use of spotting scope may be necessary to accurately survey the area. If an ATV or other vehicle is needed to cover large survey routes, the Bird Monitor is to stop at intervals of no greater than 600 feet to visually inspect for breeding activity.

b. Once breeding or nesting behavior is confirmed by the presence of a scrape, eggs or young, the Contractor (or their designee) is to notify the Contracting Officer and Project Biologist within 24 hours.

c. Report Submission: Within 30 days of completion of the project, submit the Logs/Summary of Bird Nesting Monitoring Report to the Project Biologist, Contracting Officer and cc the Environmental Compliance POC.

3.1.5.13 Shorebird Buffer Zones and Travel Corridors

The Bird Monitor(s) and Contractor are to meet the following requirements:

a. The Bird Monitor(s) are to establish a disturbance-free buffer zone around any location within the project area where shorebirds have been engaged in breeding behavior, including territory defense. A 300-foot buffer is to be established around each nest or around the perimeter of each colonial nesting area. A 300-foot buffer is to also be placed around the perimeter of areas where shorebirds are seen digging nest scrapes or defending nest territories. All construction activities, movement of vehicles, stockpiling of equipment, and pedestrian traffic are prohibited in the buffer zone. Smaller, site-specific buffers may be established if determined appropriate when reviewed in coordination with the Contracting Officer. Travel corridors are to be designated and marked outside the buffer areas for pedestrian, equipment or vehicular traffic.

b. The Bird Monitor(s) are to keep breeding sites under sufficient surveillance to determine if birds appear agitated or disturbed by construction or other activities in adjacent areas. If birds appear to be agitated or disturbed by these activities, then the Bird Monitor(s) are to immediately widen the buffer zone to a sufficient size to protect breeding birds.

c. The Bird Monitor(s) are to ensure that reasonable and traditional pedestrian access is not blocked in situations where breeding birds will tolerate pedestrian traffic. This is generally the case with lateral movement of beach-goers walking parallel to the beach at or below the highest tide line. Pedestrian traffic may also be tolerated when breeding was initiated within 300 feet of an established beach access pathway. The Bird Monitor(s) are to work with the Contracting Officer to determine if pedestrian access can be accommodated without compromising nesting success. These site-specific buffers must be determined in coordination with the Contracting officer.

d. The Bird Monitor(s) are to ensure that the perimeters of designated buffer zones are marked according to FSA Posting Guidelines (available at: https://flshorebirdalliance.org/get-involved/posting-sites/) with posts, twine and FWC-approved signs stating “Do Not Enter, Important Nesting Area” or similar language around the perimeter (see example of signage for marking designated buffer zones at http://myfwc.com/conservation/you-conserve/wildlife/shorebirds/). Posts must not exceed 3 feet in height once installed. Symbolic fencing (twine, string or rope) is to be placed between all posts at least 2.5 feet above the ground and rendered clearly visible to pedestrians. If pedestrian pathway and/or equipment travel corridor modifications are determined and coordinated with the Contracting Officer, these are to be clearly marked. Posting is to be maintained in good repair until no active nests, eggs, or flightless young are present. Although solitary nesters may leave the buffer zone temporarily with their chicks, the posted area continues to provide a potential refuge for the family until breeding is complete. Breeding is not considered to be completed until all chicks have fledged.

e. The Contractor is to ensure that no construction activities, pedestrians, moving vehicles, or stockpiled equipment are allowed within the buffer area.

f. The Contractor is to ensure that the Bird Monitor(s) designate and mark travel corridors outside the buffer areas so as not to cause disturbance to breeding birds. Heavy equipment, other vehicles, or pedestrians may go past breeding areas in these corridors. However, other activities such as stopping or turning heavy equipment and vehicles are to be prohibited within the designated travel corridors adjacent to the breeding site.

g. When flightless shorebird chicks are present within or adjacent to travel corridors, construction related vehicles are to not be driven through the corridor unless a Bird Monitor is present to adequately monitor the travel corridor. The Contractor is to avoid any chicks that may be in the path of moving vehicles. The Contractor is to level any vehicular tracks that may be capable of trapping flightless chicks, while avoiding any impacts to the chicks.

h. Notification. Any injury or death of a shorebird (including crushing eggs or young) resulting from project activities are to be reported immediately to the Contracting Officer and FWC Regional Shorebird Contact.

i. Work Delay: Delays in work due to the fault of negligence of the Contractor or the Contractor's failure to comply with this specification is to not be compensable. Any adjustments to the contract performance period or price that are required as a result of compliance with this section is to be made in accordance with the Clause SUSPENSION OF WORK of Section 00700 CONTRACT CLAUSES in Volume 1.

3.1.5.14 Protection of Piping Plover and Rufa Red Knot

a. Beginning of construction (except for May 15 to July 15) the Contractor must conduct surveys for piping plovers and red knot within the entire beach fill template project area which might support these species. Surveys should be conducted three times per month (spaced at least 9 days apart) during construction. Piping Plover and Rufa Red Knot identification, especially when in non-breeding plumage, can be difficult. The person(s) conducting the survey must demonstrate the qualifications and ability to identify shore bird species and be able to provide the information listed below. The monitor should utilize the FWC's nonbreeding shorebird data sheet forms to submit the information. If online entry becomes available, all shorebird survey data will also be entered into the FWC database online. The following is to be collected, mapped, and reported to the Contracting Officer as the Winter Shorebird Survey one week following each survey event:

1. Date, location, time of day, weather, and tide cycle when survey was conducted.

2. Latitude and longitude of observed piping plover and or rufa red knot locations (decimal degrees preferred).

3. Any color bands observed on piping plovers and rufa red knots.

4. Behavior (e.g., foraging, roosting, preening, bathing, flying, aggression, walking).

5. Landscape features(s) where piping plovers and rufa red knots are located (e.g., inlet spit, tidal creeks, shoals, lagoon shoreline).

6. Habitat features(s) used when observed (e.g., intertidal, fresh wrack, old wrack, dune, mid-beach, vegetation).

7. Substrata used (e.g., sand, mud/sand, mud, algal mat).

8. The amount and type of recreational use (e.g., people, dogs on or off leash, vehicles, kite-boarders).

9. All other shorebirds/waterbirds seen within the survey area.

10. Surveys may be conducted at various tides, but slack times (full low, full high) should generally be avoided.

11. Each month, at least one survey will be conducted in early morning (starting at sunrise). If the survey cannot be completed within three hours of sunrise, it may be completed over multiple, consecutive days.

12. Use of an ATV/ORV is possible, but access must be arranged with individual municipalities in the survey area.

13. 30 days prior to initial construction, the Contractor is to delineate and submit to the Contracting Officer the preferred piping plover and rufa red knot habitat (intertidal portions of ocean beaches, ephemeral pools, washover areas, wrack lines) outside of the project footprint but potentially impacted by the construction activity. Obvious identifiers are to be used (for example, pink flagging on metal poles) to clearly mark the beginning and end points to prevent accidental impacts to use areas.

14. Piping plover and rufa red knot habitat delineated outside of the

beach fill templates are to be avoided to the maximum extent practicable when staging equipment, establishing travel corridors, and aligning pipeline.

15. Driving on the beach for construction is to be limited to the minimum necessary within the designated travel corridor, which will be established just above or just below the primary "wrack" line.

16. Predator-proof trash receptacles are to be installed and maintained during construction at all beach access points used for the project construction to minimize the potential for attracting predators of piping plovers and rufa red knot. Workers are to be briefed on the importance of not littering and keeping the project area trash and debris free. See Appendix B to the Piping Plover Programmatic Biological Opinion. Examples of suitable receptacles <http://www.saj.usace.army.mil/About/DivisionsOffices/Planning/EnvironmentalBranch/EnvironmentalDocuments.aspx#District_Wide>.

17. Installation of ten educational signs at public access points within the project area with emphasis on the importance of the beach habitat and wrack for piping plovers and rufa red knot, locations to be submitted and approved by the Contracting Officer. When the project area has a pet or dog regulation, the provisions of the regulation are to be included on the educational signs. Digital files of the educational signs will be provided by the Contracting Officer for reproduction.

3.1.5.15 Protection of Gopher Tortoise (GT) Populations (Gopherus polyphemus)

The Contractor must keep construction activities under surveillance, management, and control to prevent impacts to GTs and their burrows. The Contractor must stay at least 25 feet from the entrance of individual burrows. All construction personnel are to be advised that GTs are listed by the State of Florida as a Threatened Species and protected by the FAC, Chapter 68A-27.004. The Contractor must be held responsible for taking, harming, or harassing the tortoises, their eggs or their burrows as a result of the construction. The destruction of GT burrows constitutes taking under this law. If a burrow cannot be avoided, the Contractor must contact the Contracting Officer and Project Biologist prior to any construction activity within 25 feet of the borrow. In addition, if a burrow cannot be avoided, the Contractor must obtain a relocation permit from FWCC and abide by the trapping and relocation permitting conditions listed in the State of Florida's "GOPHER TORTOISE PERMITTING GUIDELINES" located on the web at: http://www.myfwc.com/license/wildlife.

Taking: If the construction work kills tortoises, it will be the Contractor's responsibility to obtain an emergency take permit from the State and the FFWCC and pay the fine associated with the permit. Taking includes the entombment or killing of gopher tortoises as a result of bulldozing, grading, paving, or building construction. If such occurs, the Contractor must notify the Project Biologist.

3.1.6 Seagrass Protection Measures

a. Instruct all personnel associated with the project of the presence of seagrasses and the need to avoid contact with seagrasses.

b. All construction personnel are to advised that there are civil and criminal penalties for harming or destroying seagrasses. The Contractor may be held responsible for any seagrasses harmed or destroyed due to construction activities.

c. Do not anchor, place pipeline, or stage equipment in a manner that will cause any damage to seagrasses or hardbottoms. Anchoring, placing pipeline, or staging equipment is to avoid these sensitive areas. If such activities cannot be done without affecting these sensitive areas, the activities are to cease and the Contracting Officer and Project Biologist are to be immediately notified (no later than the morning following the next working day if the incident occurs after normal working hours). Any actual or potential incident involving damage to, or disturbance of, seagrasses or hardbottoms is to be reported.

3.1.7 Protection of Air Resources

Keep construction activities under surveillance, management, and control to minimize pollution of air resources. All activities, equipment, processes and work operated or performed by the Contractor in accomplishing the specified construction is to be in strict accordance with the applicable air pollution standards of the State of Florida (Florida Statute, Chapter 403 and others and Chapters 200 series of the FAC) and all Federal emission and performance laws and standards, including the U.S. Environmental Protection Agency's Ambient Air Quality Standards.

3.1.7.1 Particulates

Particulates, such as dust, is to be controlled at all times, including weekends, holidays, and hours when work is not in progress. Maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and work areas within or outside the project boundaries free from particulates that would cause air pollution standards to be exceeded or that would cause a hazard or nuisance. Provide equipment and approved methods to control particulates as the work proceeds and before a problem develops.

3.1.7.2 Burning

All burning is to be subject to State and local requirements, including requirements for burn permits and bans during certain conditions such as droughts.

3.1.7.3 Odors

Odors are to be controlled at all times for all construction activities.

3.1.8 Protection of Sound Intrusions

Keep construction activities under surveillance and control to minimize damage to the environment by noise.

3.1.9 Pet(s) Control

If the Contractor chooses to bring domestic pet(s) on-site, the animal(s) are to be kept within fenced areas or inside buildings unless on a leash.

3.2 POSTCONSTRUCTION CLEANUP

Clean up any area(s) used for construction.

3.3 PRESERVATION AND RESTORATION OF LANDSCAPE AND AQUATIC VEGETATION DAMAGES

Restore all landscape features and aquatic vegetation damaged or destroyed during construction operations outside the limits of the approved work areas. Such restoration is to be a part of the Environmental Protection Plan as defined in subparagraph ENVIRONMENTAL PROTECTION PLAN of paragraph SUBMITTALS above. This work is to be accomplished at the Contractor's expense.

3.4 INVASIVE AND NUISANCE SPECIES

3.4.1 Prevention of Invasive and Nuisance Species Transfer

Thoroughly clean equipment prior to and following work on the project site to ensure that items/materials including, but not limited to, soil, vegetative debris, eggs, mollusk larvae, seeds, and vegetative propagules are not transported from a previous work location to this project site, nor transported from this project site to another location. Prevention protocols require cleaning all equipment surfaces, including but not limited to, undercarriages, tires, and sheet metal. All equipment, including but not limited to, heavy equipment, vehicles, trailers, ATV's, and chippers must be cleaned. Smaller equipment, including, but not limited to, chainsaws, loppers, shovels, and backpack sprayers, must be cleaned and inspected to ensure they are free of eggs, vegetative debris, vegetative propagules, etc. Utilize any method accepted by the Government; common accepted methods include pressure washing and steam cleaning/washing equipment. Prevention protocols are to also address clothing and personal protective equipment.

Prior to the commencement of work, complete and provide an invasive and nuisance species transfer prevention plan to the Corps for approval. This plan is to be part of the Environmental Protection Plan as defined in subparagraph ENVIRONMENTAL PROTECTION PLAN above. The invasive and nuisance species transfer prevention plan is to identify specific transfer prevention procedures and designated cleaning sites/locations. Prevention protocols may vary depending upon the nature of the project site. Ensure all equipment coming onto and leaving the project site is inspected and not harboring materials that would spread, or potentially spread, invasive and nuisance species onto or off the project site. Provide a report with photographs, verifying equipment brought on site was cleaned and provide a report verifying equipment was cleaned prior to removal from the project site.

Work delays due to the fault and/or negligence of the Contractor or the Contractor's failure to comply with this specification is not to be compensable.

3.4.2 Invasive and Nuisance Species Reporting

Report sightings of invasive and nuisance species not identified and documented in the pre-construction condition within 24 hours. The reporting is to include the date, time, location (latitude and longitude preferred), environmental conditions, circumstances surrounding the sighting, disposition/behavior of the species, and any other notable observations. Pictures of the reported species are desirable. Reporting is to be made to the Jacksonville District Corps of Engineers Planning Division, Environmental Branch and Operations Division, Invasive Species Management Branch, InvasivePreventionPlan@usace.army.mil. Points of contact will be provided in the Preconstruction Conference.

3.4.2.1 Invasive And Nuisance Species Report

Within 30 calendar days following completion of work, submit a report with photographs verifying equipment brought on site was cleaned to the Contracting Office, Project Biologist, and Environmental Compliance POC. Provide a report with photographs verifying equipment was cleaned prior to removal from the project site.

3.5 MAINTENANCE OF POLLUTION CONTROL FACILITIES

Maintain all constructed facilities and pollution control facilities and devices for the duration of the contract or for that length of time construction activities create the particular pollutant.

3.6 CONSTRUCTION FORMS AND DETAILS

See the Construction Forms & Details module at the following web address: https://www.saj.usace.army.mil/About/Divisions-Offices/Engineering/Design-Branch/Specifications-Section/

Also, see web site address https://odess.usace.army.mil to obtain the Sea Turtle/Post-Hopper Dredging Project Checklist, Endangered Species Observer Program Forms, Sea Turtle Tagging and Relocation Report, and Sea Turtle Trawling Report. Links to these forms are under the heading Turtle Information.

-- End of Section --

APPROVED
By Zach Boudreau at 10:56 am, Jul 30, 2021

Biological Monitoring Plan

St. Lucie County South Beach Restoration

Permit No. 0154626-001-JC

July 2021

Table of Contents

1.0 INTRODUCTION

The South St. Lucie County Beach and Dune Restoration Project (Project) authorized by Florida Department of Environmental Protection (FDEP) Permit Number 0154626-001-JC and U.S. Army Corps of Engineers (USACE) permit number SAJ-2009-03448 (IP-GGL) includes the restoration and periodic nourishment of approximately 3.4 miles of beach and dune between FDEP reference monuments R-98 in St. Lucie County and R-1 in Martin County. In addition to upland sand mines, offshore sand sources (borrow areas) and pipeline corridors are authorized for use.

Hardbottom habitat parallels the shoreline along South Hutchinson Island and is found between R-99.5 and R-115 within the project area (also updrift and downdrift of the project). Within the project area, hardbottom tends to be closer to shore in the north than in the south. The habitat consists of emergent Anastasia coquina limestone and features typical “ledge” relief. Relief features facilitate the formation of diverse communities, particularly in areas with higher relief (tall ledges) or those farther offshore that tend be exposed for longer durations (often permanently). The Project is predicted to impact approximately 0.57 acres of nearshore hardbottom from Project fill and pipeline corridor use. 0.97 acres of mitigative reef is required to be constructed to offset permitted impacts to hardbottom resources.

Permit No. 0154626-001-JC does not authorize any direct and / or secondary (indirect) project related impacts to hardbottom resources beyond those permitted and requires that biological monitoring be conducted to provide characterization of the post-construction impacts as well as biotic response to the project via monitoring of nearshore hardbottom, marine turtles (nesting and in-water), shorebirds, and dune plantings. Any unpermitted direct and / or secondary adverse impacts to nearshore hardbottom resources shall be documented and offset if they occur. Further, biological monitoring is required to document the success of the mitigation reef in meeting its specified success criteria. This document outlines the protocols for the Project Biological Monitoring Plan based on the issued FDEP permit, the USACE permit, National Marine Fisheries Service (NMFS) Conservation Recommendations, and U.S. Fish & Wildlife Service (USFWS) Statewide Programmatic Biological Opinion (SPBO).

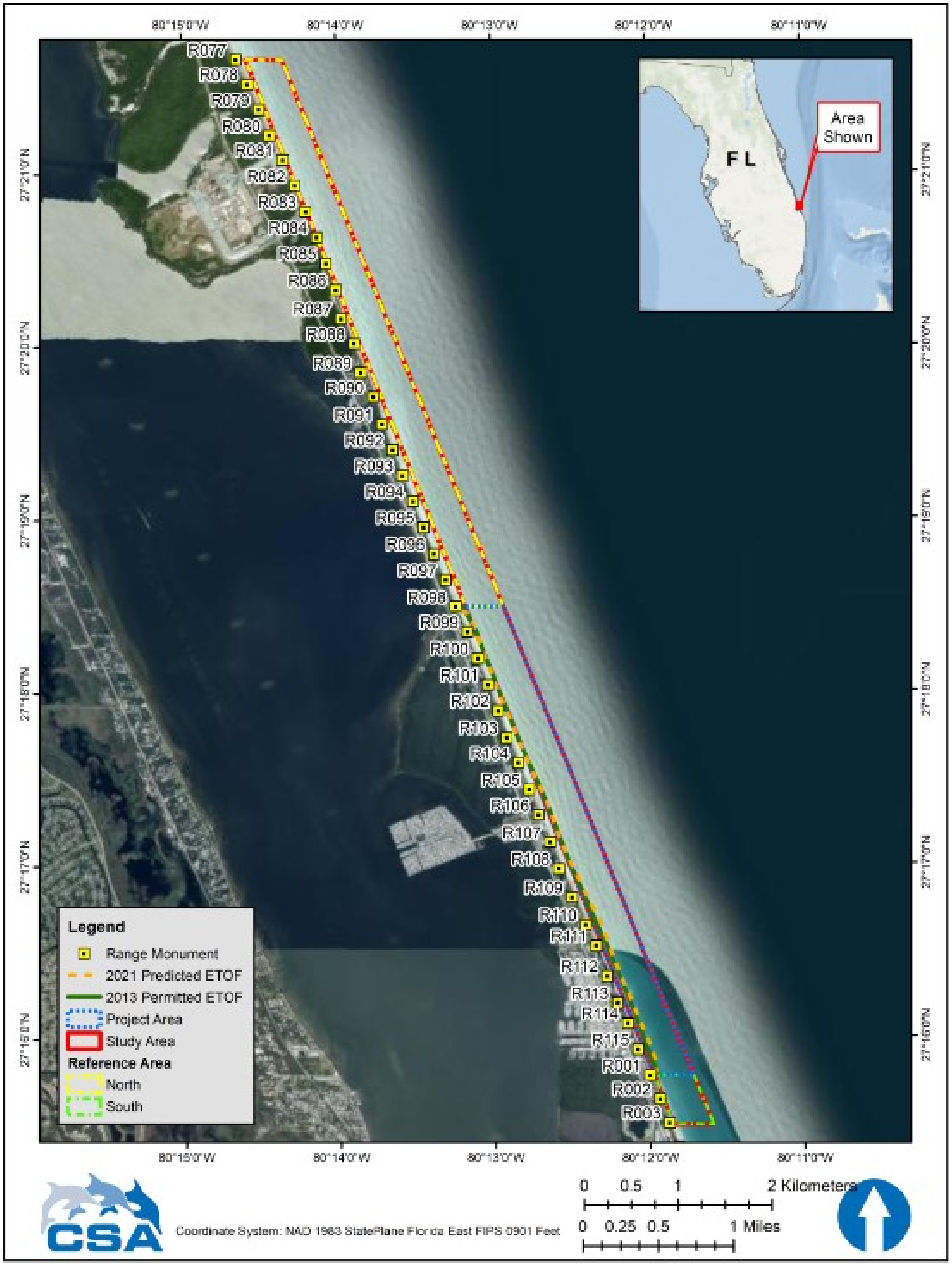

Figure 1. South St. Lucie County Project Area: FDEP reference monuments R-98 to the St. Lucie/Martin County line.

2.0 NEARSHORE HARDBOTTOM MONITORING

Nearshore hardbottom monitoring includes surveys in the nearshore region within the influence of the permitted Equilibrated Toe of Fill (ETOF). Hardbottom edge mapping and transect surveys (**Section 2.1**) and hardbottom side scan sonar surveys (**Section 2.2**) are required.

2.1 Nearshore Hardbottom Monitoring

Biological monitoring for beach fill placement under this permit shall include a pre-construction (baseline) monitoring event during the summer prior to the initial construction event, an initial post-construction monitoring event (within six months of project completion), and three annual post-construction monitoring events (Years 1, 2, and 3 post-construction). Each subsequent nourishment conducted under this permit will initiate a complete round of post-construction monitoring (i.e., initial post-construction event and three annual post-construction events). The pre-construction monitoring event conducted in August of 2012 prior to the initial nourishment shall serve as the baseline for all subsequent monitoring events. Nearshore hardbottom updrift, adjacent to, and downdrift of the fill template (but outside of the ETOF) shall be monitored. All monitoring events shall be conducted and completed during summer months (May 1 through September 30).

During the summer monitoring season, weather and tide conditions will be monitored for an appropriate window of opportunity (acceptable water clarity and visibility) for performance of the fieldwork, and a daily log of weather and tide conditions will be maintained at minimum from May 1 to September 30 of each monitoring year. When an appropriate window occurs, personnel and equipment will be mobilized to conduct the fieldwork. Standard operating procedures shall be used during each monitoring event to provide consistent and repeatable collection of data. The aim of biological monitoring is to identify any unpermitted direct and / or secondary adverse impacts to nearshore hardbottom resources due to the spreading of project sand farther than permitted (i.e., seaward of the permitted ETOF). As such, surveys conducted during each monitoring event will document sediment depth and cover as well as the abundance, distribution, condition, and function of hardbottom resources (biotic assemblages) within areas potentially under the influence of the project (**Figures 2, 3,** and **4**). Hardbottom monitoring will consist of nearshore hardbottom edge mapping (**Section 2.1.1**), transect surveys (**Section 2.1.2**), and sonar surveys (**Section 2.2**) as described below.

Figure 2. Nearshore hardbottom monitoring transects established within the South St. Lucie County North Reference Area.

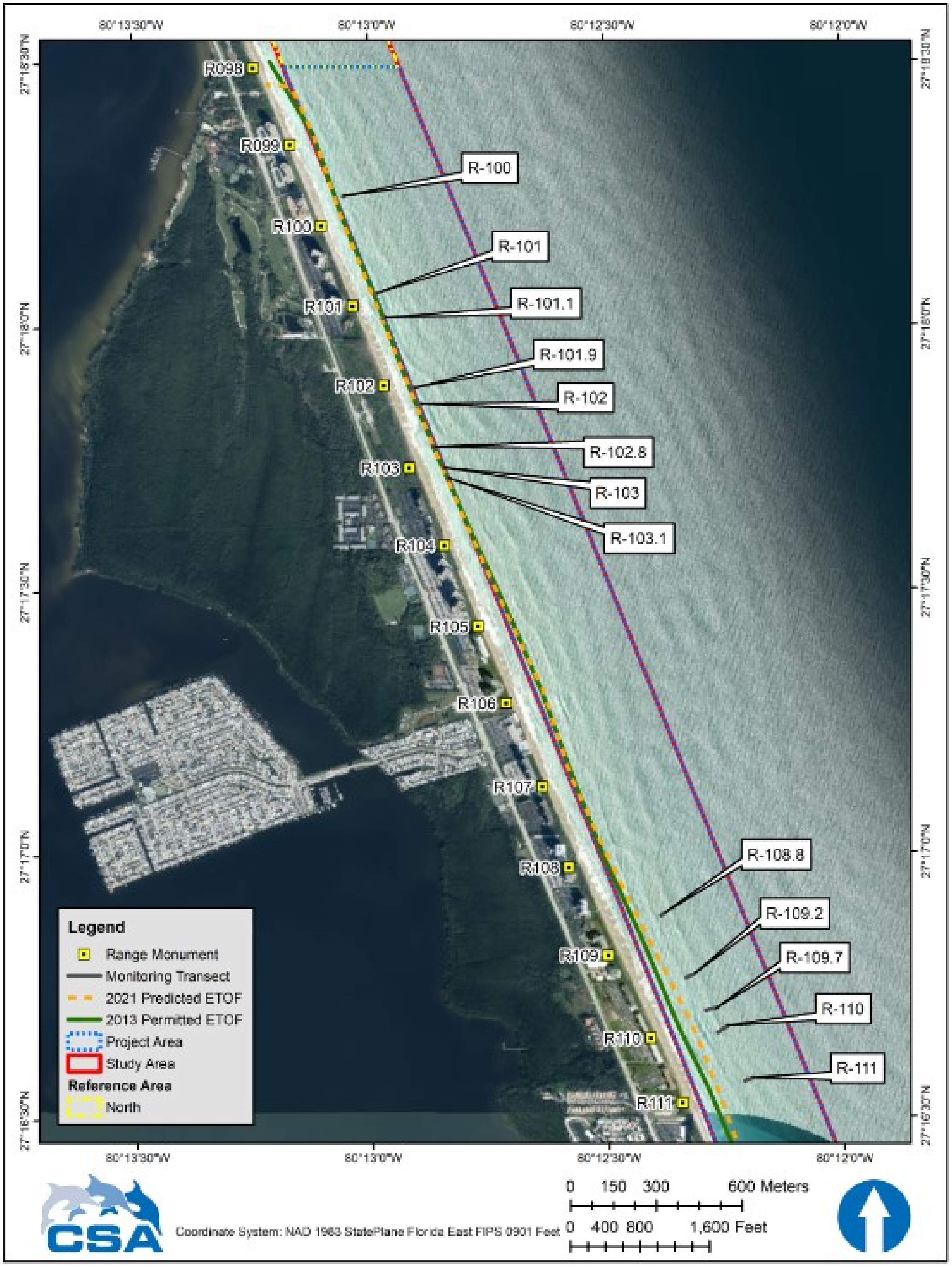

Figure 3. Nearshore hardbottom monitoring transects established within the South St. Lucie County Project Area.

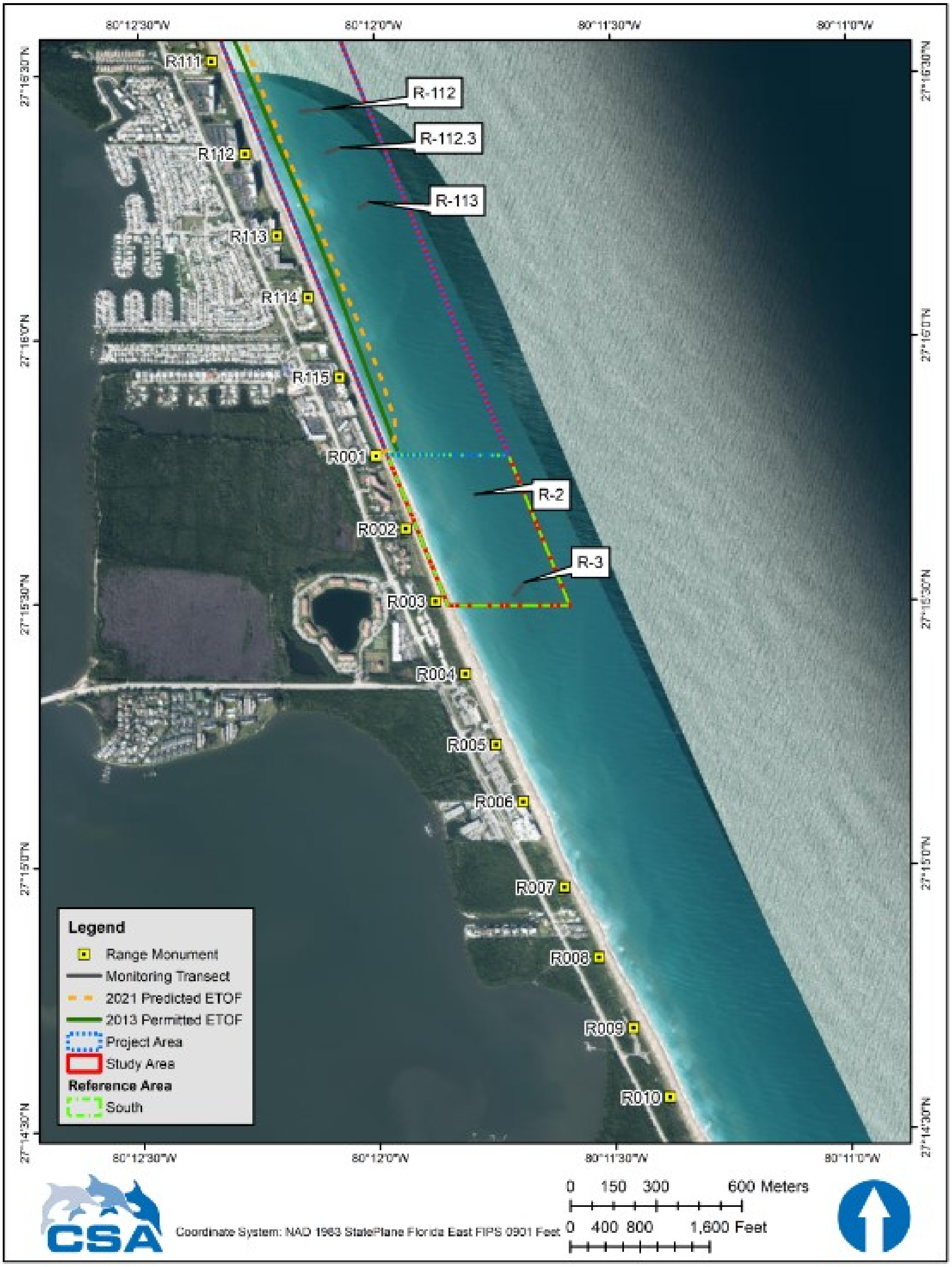

Figure 4. Nearshore hardbottom monitoring transects established within the South St. Lucie County Project Area and South Reference Area.

2.1.1 Nearshore Hardbottom Edge Mapping

In situ delineation of the nearshore hardbottom edge will provide information on hardbottom exposure within the project area and allow for determination of direct impacts, when occurring, due to hardbottom burial by project fill. The entire length of the nearshore hardbottom edge between FDEP reference monuments R-78 and R-92.5 and between R-97 (St. Lucie County) and R-3 (Martin County) shall be mapped during each monitoring event (**Figures 2, 3,** and **4**). The nearshore hardbottom edge is defined as the visible border between sand and hardbottom. Bounce dives shall not be used to delineate the hardbottom edge unless authorization is granted by FDEP staff. If necessary, requests will be submitted to resource review staff in the Department's Beaches, Inlets, and Ports program.

For in situ hardbottom edge mapping, at least two divers shall, together, swim the entire length of the hardbottom edge. One diver shall tow a DGPS antenna transmitting continuous positions to HYPACK hydrographic survey software on board a survey vessel. To accurately map the edge, the towing-diver shall swim at a speed conducive to maintaining the buoy on as short a tether as possible. The non-towing diver shall record qualitative digital video to document the nearshore hardbottom edge for descriptive analysis (e.g., of the dominant benthic communities, vertical relief, and sand cover). To allow for visual characterization of the hardbottom edge, the recording diver shall position the camera at an oblique angle to the seafloor. Positions of breaks (sand gaps) in the hardbottom edge greater than 5 meters in length will be noted during the survey. If possible, all video shall be georeferenced. The final deliverable video shall include survey date and location coordinates superimposed.

2.1.2 Establishment and Monitoring of Permanent Transects

Hardbottom monitoring updrift, adjacent to, and downdrift of the fill template (outside of the permitted ETOF) will occur along permanent shore-perpendicular transects. Line-intercept surveys, interval sediment depth measurements, quadrat sampling, and digital video collection will be conducted along each transect during each monitoring event. To obtain the most accurate information on sediment depth and the location of sediment and hardbottom, line-intercept and interval sediment depth surveys shall be conducted first along each transect during each event.

2.1.2.1 Transect Establishment

A total of 26 permanent, shore-perpendicular monitoring transects shall be surveyed for this project during each monitoring event (**Table 1** and **Figures 2, 3,** and **4**). Sixteen (16) of these transects are within the project area (i.e., nearshore area under the influence of the project) while the remaining 10 are either updrift in the northern reference area (N=8 transects) or downdrift in the southern reference area (N=2 transects [a and b at R-2 considered one transect]) (**Table 1**).

Table 1. Areas and transect names, installation dates, and locations for the St. Lucie South Beach Nourishment Project. The 16 project area transects are highlighted in light pink.

Area	Transect					
	Name	Installation	Location			
			Start		End	
			Latitude	Longitude	Latitude	Longitude
		(Date)	(DMS)	(DMS)	(DMS)	(DMS)
North	**R-78.7**	7/2/2009	27° 21' 25.0135" N	80° 14' 22.4675" W	27° 21' 25.2200" N	80° 14' 19.8560" W
Reference	**R-80.4**	6/30/2009	27° 21' 09.2286" N	80° 14' 20.1454" W	27° 21' 09.5550" N	80° 14' 14.7070" W
	R-82.8	7/1/2009	27° 20' 48.5043" N	80° 14' 10.3383" W	27° 20' 48.6970" N	80° 14' 08.4620" W
	R-85	7/3/2009	27° 20' 28.5472" N	80° 14' 01.8910" W	27° 20' 29.1760" N	80° 13’ 58.2890" W
	R-86.3*	7/9/2009	27° 20' 16.2581" N	80° 13' 57.7805" W	27° 20' 16.6730" N	80° 13’ 53.0010" W
	R-88.7	7/10/2009	27° 19' 53.7581" N	80° 13' 48.1805" W	27° 29’ 53.1220" N	80° 13’ 44.8400" W
	R-90.4*	6/30/2009	27° 19' 38.9291" N	80° 13' 42.2675" W	27° 19’ 39.3800" N	80° 13’ 39.4120" W
	R-92*	6/29/2009	27° 19' 23.5721" N	80° 13' 36.6064" W	27° 19’ 24.0780" N	80° 13’ 34.2380" W
Project	**R-100**	6/28/2009	27° 18' 15.0222" N	80° 13' 03.5404" W	27° 18' 16.5737" N	80° 12' 58.1109" W
	R-101	7/15/2009	27° 18' 03.7127" N	80° 12' 59.2210" W	27° 18' 05.4209" N	80° 12' 54.1120" W
	R-101.1	8/19/2012	27° 18' 00.8922" N	80° 12' 58.2545" W	27° 18' 02.5338" N	80° 12' 53.1465" W
	R-101.9	8/19/2012	27° 17' 52.8522" N	80° 12' 54.7747" W	27° 17' 54.5153" N	80° 12' 49.6291" W
	R-102	6/28/2009	27° 17' 51.4961" N	80° 12' 54.2525" W	27° 17' 52.7642" N	80° 12' 48.7682" W
	R-102.8	8/19/2012	27° 17' 46.2941" N	80° 12' 51.8944" W	27° 17' 47.9304" N	80° 12' 46.7856" W
	R-103	8/16/2012	27° 17' 43.8521" N	80° 12' 50.6646" W	27° 17' 45.6611" N	80° 12' 45.5243" W
	R-103.1	8/19/2012	27° 17' 42.6761" N	80° 12' 50.1547" W	27° 17' 44.3303" N	80° 12' 44.9916" W
	R-108.8	8/17/2012	27° 16' 52.9721" N	80° 12' 23.2023" W	27° 16' 54.5142" N	80° 12' 17.7906" W
	R-109.2	7/10/2009	27° 16' 46.1023" N	80° 12' 20.2685" W	27° 16' 47.9898" N	80° 12' 15.2311" W
	R-109.7	8/20/2012	27° 16' 42.1002" N	80° 12' 17.6765" W	27° 16' 43.5976" N	80° 12' 12.4599" W
	R-110	7/2/2009	27° 16' 39.5202" N	80° 12' 15.9965" W	27° 16' 41.6200" N	80° 12' 10.9387" W
	R-111**	7/15/2009	27° 16' 34.0060" N	80° 12' 12.7746" W	27° 16' 36.2481" N	80° 12' 07.8701" W
	R-112	6/29/2009	27° 16' 25.8763" N	80° 12' 09.3872" W	27° 16' 26.5060" N	80° 12' 03.8824" W
	R-112.3	8/20/2012	27° 16' 21.1541" N	80° 12' 06.1567" W	27° 16' 22.8020" N	80° 12' 01.0282" W
	R-113	8/21/2012	27° 16' 14.9263" N	80° 12' 01.9324" W	27° 16' 16.7334" N	80° 11' 56.8626" W
South	**R-2 (a+b) ŧ**	8/19/2012	27° 15' 42.2109" N	80° 11’ 47.5591" W	27° 15' 42.2590" N	80° 11' 47.4240" W
Reference	**R-3**	8/17/2012	27° 15' 30.6461" N	80° 11' 42.6246" W	27° 15' 32.1970" N	80° 11' 41.2740" W

* Northern reference area transects R-86.3, R-90.4, and R-92 were added to the survey requirements late in the 2012 monitoring season and, due to time and weather, were not monitored until the summer of 2013.

** The 2009 Transect R-110.5 was surveyed in 2012 and determined to be located closer to R-111.

ŧ South reference area transect R-2 was installed as two parallel 4-m transects (transects R-2 a and R-2b) due to the small amount of hardbottom present.

Twenty-three of the transects were surveyed during the pre-construction baseline monitoring event in 2012 while the other three (3) transects (transects 86.5, 90.4, and 92 in the northern reference area) were added in 2013 (see **Table 1**). If site conditions allowed (hardbottom was present), transects up to 150-m long (maximum length) were established during the pre-construction baseline monitoring event. The length of each transect was determined by the cross-shore distance from the nearshore to the offshore hardbottom edge during the baseline event. Transect positions and lengths are permanent. Thus, once established during the baseline monitoring event, the same transects in the same positions and of at least the same lengths must be surveyed in all subsequent monitoring events. To ensure repeatability in transect placement during monitoring events, permanent markers (pins, iron rods, etc.) shall be installed at set points along each transect at the time of establishment.

2.1.2.2 Quadrat Establishment

Each 150 m transect shall have 12, 0.25 m^2 quadrats within the first 125 meters, with the first one being located along the nearshore hardbottom edge. Fewer quadrats may be employed along shorter transects (**Table 2**).

Table 2. Areas, transects, and number and positions of quadrats for South Beach Nourishment.

Area	Transect	No. of Quads (n)	Position of Quadrats (m)												
North Reference	**R-78.7**	8	0	5	10	19	30	37	67	75					
	R-80.4	7	0	5	10	15	23	30	36						
	R-82.8	7	0	5	10	20	30	37	43						
	R-85	13	0	5	12	20	30	40	51	60	70	81	93	100	102
	R-86.3	13	0	5	11	20	29	40	50	60	70	80	89	101	122
	R-88.7	12	0	5	10	20	31	39	50	60	70	81	92	95	
	R-90.4	8	0	12	20	30	50	60	70	80					
	R-92	9	0	5	10	20	30	32	35	38	40				
Project	**R-100**	4	0	2	5	6.3									
	R-101	6	0	3	4	5	7	9.3							
	R-101.1	6	0	1.7	6	9.7	12	15							
	R-101.9	6	0	2	4	6.3	12	17							
	R-102	7	0	2	5	7.3	8.5	10	14*						
	R-102.8	4	0	3.3	5	8									
	R-103	4	0	1.6	4.3	5									
	R-103.1	4	0	1.7	5	9									
	R-108.8	5	0	3	5	6.3	18								
	R-109.2	6	0	1	5**	10	30	35							
	R-109.7	6	0	4	6.5	35	37.6	40							
	R-110	10	0	1	3.8	8.8	10	17	28	30	34.5	42			
	R-111	6	0	1.5	8	16	19[#]	31[##]							
	R-112	8	0	32.8	34	42	50	59	72	75					
	R-112.3	9	0	4	10	15	18	25	35	42	51				
	R-113	8	0	3	7.5	12	22	33	40						
South Reference	**R-2a**	2	0	4											
	R-2b	2	0	4											
	R-3	8	0	5	10	17	43	47.5	54	57					

* Initially (2012) recorded as meter 13.5. ** Initially (2012) recorded as meter 4.5.
[#] Initially (2012) recorded as meter 20. [##] Initially recorded as meter 32.

During establishment, quadrats shall be positioned such that areas covered by sand are avoided (i.e., quadrat placement during establishment will be biased to include hardbottom). Thus, quadrats shall not be placed in sand gaps between hardbottom ridges. All quadrats are to be permanent. Once established during the pre-construction (baseline) monitoring event, the same number of quadrats in the same positions as in the baseline shall be sampled in all subsequent monitoring events regardless of the exposure or burial condition of each location after initial quadrat establishment. To ensure repeatability in quadrat placement during monitoring events, pins (or nails or eyebolts) shall be installed to permanently mark the location of each quadrat. The permanent location of each quadrat shall be recorded and reported for each monitoring event and post-construction monitoring event quadrat positions shall match baseline quadrat positions.

2.1.2.3 Transect Monitoring

2.1.2.3.1 Line-Intercept Survey

In order to document larger areas of uninterrupted sand (physical transitions along the monitoring transects between sand and hardbottom), and to track changes in sediment cover on the hardbottom, line-intercept surveys will be conducted along each permanent transect (biological and sediment only). During each monitoring event, the landward and seaward position of each sand patch / trough at least 0.5 m in length shall be recorded along each transect by reference to transect tape meter marks. Meter mark references shall be to one decimal place (e.g., patch from 2.4 to 3.2 m).

2.1.2.3.2 Interval Sediment Depth Measurements

In order to track changes in sediment depth associated with changes in sediment cover, each monitoring event will include collection of interval sediment depth measurements along each permanent transect (biological and sediment only). Sediment depth shall be measured at 1-m intervals along the entire length of each transect, inclusive of sand patches. For each measurement, a ruler graduated in centimeters (0 to 30 cm) shall be pressed through the sediment until the ruler reaches the surface of hard substrata or is totally immersed in sand. Depth measurements shall be rounded to the nearest cm (i.e., sediment thickness of less than 0.5 cm will be recorded as “0 cm”, while thickness greater than 0.5 cm but equal or less than 1 cm shall be recorded as “1 cm”, etc.). Measurements greater than 30 cm will be recorded as “> 30 cm”.

2.1.2.3.3 Quadrat Sampling

Benthic communities and their habitats will be characterized quantitatively using the quadrat method, which includes sampling habitat and assemblages within permanently positioned quadrats along each biological transect. This method ensures that the same quadrats (same location, same size) are sampled in each monitoring event in order to document changes in sediment and benthic communities over time. The sampling protocol is similar to that used in the Benthic Ecological Assessment for Marginal Reefs (BEAMR) (Lybolt and Baron, 2006). As described below, three main benthic characteristics shall be assessed in each quadrat during sampling: physical structure, planar cover of sessile benthos, and coral (scleractinian and octocoral) size and density. As with all non-consumptive surveys, BEAMR is necessarily constrained to visually conspicuous organisms with well-defined, discriminating characteristics for identification.

Physical structure: Maximum topographic relief and mean sediment depth (average of three depth measurements) shall be measured (cm) within each quadrat to document physical structure.

Cover (percent) of functional groups: The distribution of substrata and composition of the benthic community within each quadrat shall be documented by estimating the planar cover (percent) of functional groups. Specifically, the cover of the following 17 functional groups shall be estimated: sediment (sand, shell-hash, or mud as subcategories), bare hardbottom, rubble, cyanobacteria, macroalgae, encrusting red algae (CCA), turf algae, sponge, octocoral, scleractinian coral, anemone, hydroid, wormrock (*Phragmatopoma* spp.), barnacle, bryozoan, echinoderm (crinoids only), and tunicate. Each functional group shall be assigned a cover value (percent) from 0% to 100%, with the total of all functional groups in each quadrat equaling 100%. Macroalgae with at least 1% cover shall be identified to genus and the cover (percent) of each genus shall be recorded. Maximum and average algal height of the two dominant macroalgal species shall also be recorded, and the average algal height shall be estimated by five measurements. Unattached or floating macroalgae shall be disregarded and shall be removed from quadrats prior to sampling. The presence of bio-eroding sponges will be noted and species/genera (e.g., *Pione lampa*, *Cliona deletrix*, *C. varians*) having at least 1% cover shall be quantified.

Coral size and density: Monitoring staff shall also measure and record to the nearest centimeter (cm) the maximum dimension (height or width) of each scleractinian coral and octocoral colony within each quadrat. The smallest size recorded shall be one (1) cm; for colonies less than one (1) cm in size, the measurement recorded shall be "< 1 cm". Each colony within each quadrat shall also be enumerated and identified (by species for scleractinians, by genus for octocorals) to determine coral density and composition.

2.1.2.3.4 Qualitative Video Recording

Qualitative video survey data collected as part of beach nourishment project biological monitoring functions as an archival data set that can be used for general reference purposes or to help resolve potential impacts suggested by quadrat and sediment survey data. As such, video data could be reviewed and compared between surveys in order to document qualitative changes along transects over time for the purpose of refining impact area assessment. Qualitative video surveys shall be conducted along the entire length of each transect at each location during each monitoring event using a digital video camera. Video of the seafloor along each transect shall progress no faster than 5 meters/minute over hardbottom and 10 m per minute over large sand patches. During the survey, a convergent laser guidance system shall be employed to precisely maintain a constant camera distance of 25 cm above the bottom. The transect line shall be clearly visible in all video so that locations may be accurately referenced. If the diver is moved off the transect by surge, the diver shall return to the point where he/she was disturbed by the wave action, and resume filming at that point.

The video shall be reviewed during the course of the survey to ensure that there are no gaps in the data due to diver error and that the quality of the video is acceptable for video analysis. Any missing video transect data or poor-quality video shall be re- filmed during the course of the event. A 360° panoramic view at an angle of roughly -30° to the horizon shall be recorded both at the beginning and end of each transect from an elevation of roughly 1 m above the bottom. At the beginning and end of each transect, a standard underwater display shall also be recorded and integrated directly into the digital video track. The standard display shall report: 1) the project FDEP permit number (e.g., 0555555-001-JC); 2) the transect number; 3) the survey date (e.g., 06/25/2021); 4) the water depth in meters for both the beginning (transect meter 0) and end (final meter) of the transect (e.g., start depth = 2 m, end depth = 4.5 m); and 5) any pertinent notes (e.g., poor visibility, large swell, etc.). Video data (files) will be supplied to FEDP during raw data submittal.

2.2 Sonar Survey of Hardbottom

Side scan sonar survey data will be collected each monitoring event in which nearshore hardbottom edge and transect surveys are required to be conducted (see **Section 2.1**). The survey area shall extend 350 m offshore of the current beach, 1,000 m updrift of the project boundary, and 3,000 m downdrift of the project boundary. Hardbottom features (especially edges) identified in the sonar surveys shall be verified in situ by qualified scientific divers using survey grade differential GPS (DGPS). Except for the nearshore hardbottom edge, verification need not be continuous. A GIS-based hardbottom habitat map (ESRI shape files) shall be generated following each survey.

3.0 PIPELINE CORRIDOR MONITORING

<u>Pre-construction pipeline corridor surveys are required each construction event (fill placement) in which an offshore borrow area will be used as a sand source</u>. This requirement provides the FDEP with reasonable assurance that impacts to hardbottom resources within pipeline corridors will be avoided or minimized to the greatest extent practicable. Pre-construction surveys (**Section 3.1**) are required for all pipeline corridors. Planned impact avoidance and minimization (**Section 3.2**) as well as post-placement pre-pumping pipeline surveys (**Section 3.3**) are only required for pipeline corridors containing hardbottom resources that will be used for a construction event. Hardbottom

monitoring within pipeline corridors is only required when resources cannot be avoided during construction (**Section 3.4**).

3.1 In-water Diver Surveys

Prior to each construction event, in situ diver surveys shall be conducted and used to identify areas of hardbottom (habitat or resources) within each pipeline corridor proposed/selected for use. During the survey, divers shall investigate the full area (length and width) of each corridor and determine the presence or absence of hardbottom resources. If resources are present, divers shall map their boundary in situ using DGPS. In addition to mapping, divers shall also note the general condition of the benthic community, identify dominant benthic organisms and substratum types, and estimate the vertical relief of hardbottom. Video shall be collected during the survey to document findings.

3.2 Impact Avoidance and Minimization

Impacts to hardbottom resources due to construction activities shall be avoided to the greatest extent practicable. If hardbottom resources cannot be avoided during construction, then impacts shall be minimized to the greatest extent practicable. Results from the current pre-construction pipeline corridor survey (**Section 2.3.1**) shall be used to determine the least impactful placement for each pipeline within each corridor for the current construction (nourishment) event. When and where necessary (i.e., if hardbottom resources cannot practicably be avoided), results of the current pre-construction survey shall be used to determine the locations along each pipeline within each corridor where collars or risers will be used to limit impacts to resources. Following analysis of survey results by the Permittee, a written description of the methods that will be used to avoid impacts to hardbottom resources shall be included in the Pre-Construction Pipeline Corridor Survey Report (**Section 5.3.2**) provided to the FDEP. If hardbottom resources cannot practicably be avoided, then the report shall also include a written description of the methods that will be used to minimize impacts to hardbottom resources and the locations in which these methods will be employed.

3.3 Post-Placement Pre-Pumping Pipeline Surveys

A post-placement, pre-pumping pipeline survey shall be conducted within each pipeline corridor identified as currently containing hardbottom resources that will be used in a construction event. Surveys may be conducted on a corridor by corridor basis (i.e., as corridors are needed/used). For each corridor, the survey shall be conducted immediately following pipeline placement, shall be conducted using survey-grade DGPS, and shall document the location of the full length of the placed pipeline along the limits of hardbottom mapped in the most recent (up to date) in situ diver corridor survey (**Section 3.1**). For each post-placement, pre-pumping pipeline survey conducted, data shall be provided to the FDEP at least 72 hours prior to the intended or actual start of pumping activities (**Section 7.3.3**). Survey results will determine the type of monitoring required for each area containing resources within 25 m to either side of the placed pipeline (**Section 3.4**).

3.4 Pipeline Corridor Monitoring

For each fill placement event, whenever hardbottom resources are present within 25 m to either side of a placed pipeline, monitoring is necessary to document potential project-related impacts, such as damage, burial, and excessive sedimentation and/or turbidity. For each pipeline corridor containing hardbottom resources, the type of monitoring required for each hardbottom patch/feature depends on whether the pipeline runs across/through or adjacent to hardbottom resources. Monitoring methods for each of these scenarios (Monitoring Types 1 and 2 [**Table 2** and **Figure 5**]) as well as actions required to be taken if impacts are documented (see **Section 3.4.4**) are described below.

Table 2. Monitoring Types and hardbottom patches/features they are required for.

Monitoring Type	Area Required
1	Areas where the pipeline runs across/through hardbottom resources
2	Areas where the pipeline runs adjacent to hardbottom resources that are within 25 m of the placed pipeline

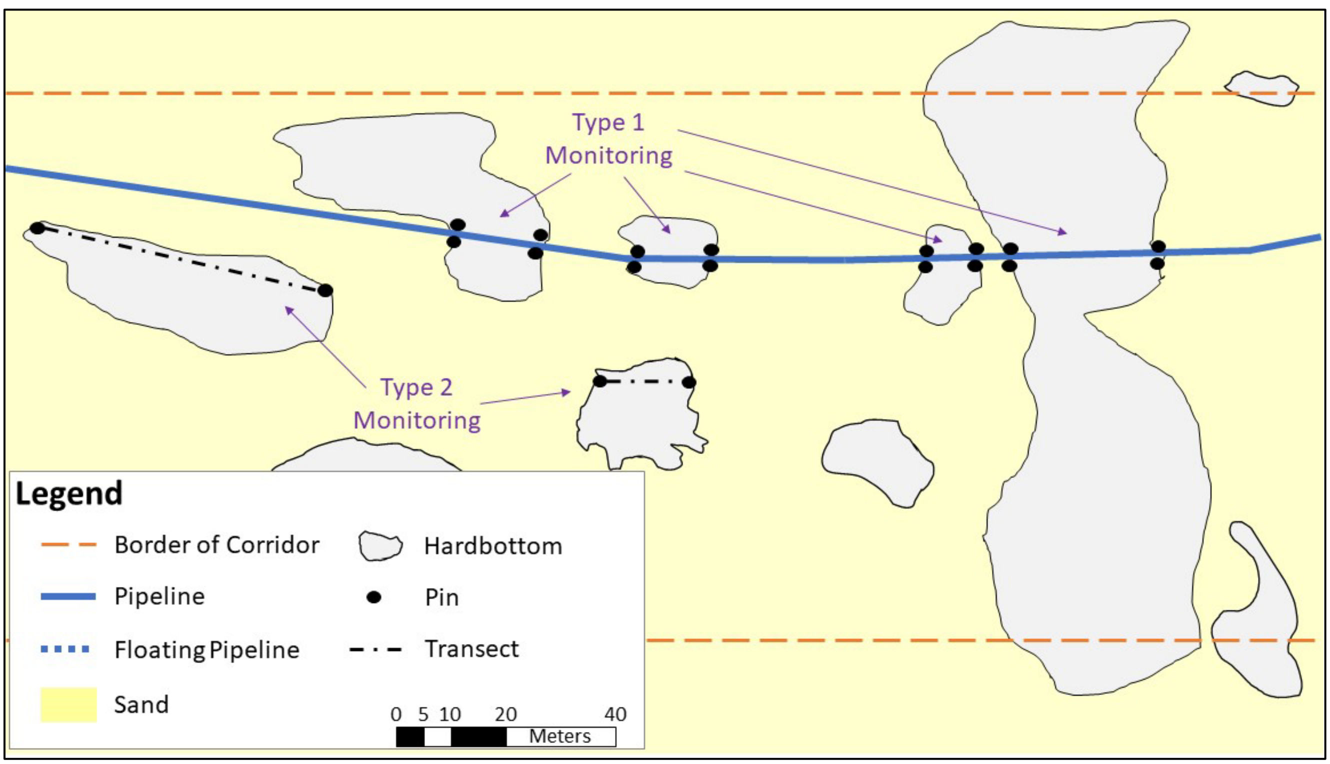

Figure 5. Depiction of pipeline corridor with placed pipeline (non-floating), hardbottom patches/features, and monitoring equipment. Monitoring pin and transect configurations for Monitoring Types 1-2 are indicated.

3.4.1 Type 1 Monitoring – Pipeline Runs Through/Across Hardbottom

Type 1 monitoring shall be conducted for each hardbottom patch/feature where the pipeline runs across/through hardbottom resources. Collars or risers shall be employed to limit impacts when pipelines run across/through hardbottom. Each area containing hardbottom resources that the pipeline runs across/through shall be monitored prior to pumping, during construction, and following construction, as described below (**Figure 5**).

The pre-pumping monitoring event shall occur as soon as possible after the pipeline has been placed/installed and shall be completed prior to any pumping activities. Initial monitoring activities for each hardbottom patch/feature shall include the establishment of pins (on both sides of the placed pipeline) where the pipeline starts to cross/run through hardbottom resources and where it ends. The coordinates (DGPS) of each pin shall be recorded. Once pins have been installed and coordinates collected, a diver with a digital, underwater video camera shall swim the length of each side of the pipeline (down one side and up the other) in each area where the pipeline crosses/runs through hardbottom and record video of the seafloor. During the pre-pumping monitoring event, a second diver shall accompany the videographer and record the path traveled during video collection and the location (coordinates) of risers/collars using survey grade DGPS. The diver conducting the video survey shall swim at a speed of approximately 5 meters/minute, shall maintain the video camera at a distance of roughly 1 m above the seafloor, and shall hold the camera at an oblique angle to the seafloor to allow for visual characterization of the area. If the diver is moved off course by waves or currents, the diver shall return to the point where he/she was disturbed and resume filming. Video shall be reviewed at the end of the survey to ensure the quality is acceptable for general characterization of the benthos; poor quality video shall be re-filmed.

During-construction monitoring shall start immediately (within 24 hours) following the initiation of pumping activities, and monitoring shall be conducted once per week until the pipeline is demobilized. For each weekly during-construction monitoring event, a diver shall collect video using the same methods employed for the pre-pumping video survey (see above). If, during construction, a pipeline leak is observed (by divers during surveys or by the dredging / pumping crew), turbidity measurements shall immediately be taken. There is no mixing zone for pipeline corridors, so turbidity shall be measured at the source of sedimentation (leak site), and background turbidity shall be measured 300 m up current of the leak. Substantial leaks are those that result in compliance turbidity measurements that exceed state water quality standards of 29 NTUs above background or that result in visual deposition of material (i.e., sedimentation on benthic resources). All dredging / pumping / filling operations shall cease immediately if a substantial leak is identified. All dredging / pumping / filling operations shall also cease immediately if impacts to hardbottom resources are observed. The JCP Compliance Officer shall be notified within 24 hours of documenting / observing substantial leaks or sedimentation or impacts to hardbottom resources and the cause of impacts shall be identified and corrected. See **Section 3.4.3** for actions required to be taken if impacts to hardbottom resources are documented during any during-construction monitoring events.

The post-construction monitoring event shall occur immediately following removal of the pipeline. For the post-construction monitoring event, a video survey shall be conducted along with a visual/video inspection of potential impact sites. Areas where video was recorded during the pre-pumping and during-construction monitoring events (i.e., seafloor along both sides of the pipeline) shall once again be surveyed using the same methods employed during previous monitoring events. Since the pipeline will have been removed, the video survey shall be conducted between the installed pins along the GPS track recorded during the pre-pumping monitoring event. Additionally, sites where the pipeline rested directly on hardbottom or hardbottom areas (sites) that were under risers or collars shall be visually inspected and videoed during the post-construction survey. GPS coordinates of these sites collected during the pre-pumping survey shall aid in this effort. Any impacts observed during the video survey or during the visual/video inspection of potential impact sites will be documented and reported to the JCP Compliance Officer within 24 hours. See **Section 3.4.3** for actions required to be taken if impacts to hardbottom resources are documented during the post-construction monitoring event.

3.4.2 Type 2 Monitoring – Pipeline Runs Adjacent to Hardbottom

Type 2 monitoring shall be conducted for each hardbottom patch/feature where the pipeline runs adjacent to hardbottom resources that are within 25 m of the placed pipeline (**Figure 5**). Each area containing hardbottom resources that the pipeline runs adjacent to shall be monitored prior to pumping and following construction as described below.

The pre-pumping monitoring event shall occur as soon as possible after the pipeline has been placed/installed and shall be completed prior to any pumping activities. Initial monitoring activities shall include the installation of a permanent transect through each area (patch/feature) containing hardbottom resources that is adjacent to, and within 25 m of, the placed pipeline. Each installed transect shall run the length of the hardbottom patch/feature adjacent to the placed pipeline, shall be oriented parallel to the pipeline, and shall be within 25 m of the pipeline. Pins or rods shall mark the start and end of each transect and coordinates (DGPS) of transect start and end points shall be recorded. Once transects have been established, a diver with a digital, underwater video camera shall swim the length of each transect and record video of the seafloor. The diver shall swim at a speed of approximately 5 meters/minute, shall maintain the video camera at a distance of roughly 1 m above the seafloor, and shall hold the camera at an oblique angle to the seafloor to allow for visual characterization of the area. If the diver is moved off course by waves or currents, the diver shall return to the point where he/she was disturbed and resume filming. Video shall be reviewed at the end of the survey to ensure the quality is acceptable for general characterization of the benthos; poor quality video shall be re-filmed. The post-construction monitoring event shall occur immediately following demobilization of the pipeline. For the post-construction event, a diver shall swim the length of each transect line and collect video as in the pre-pumping monitoring event (i.e., same methods shall be used). Any impacts observed shall be documented and reported to the JCP Compliance Officer within 24 hours. See **Section 3.4.3** for actions required to be taken if impacts to hardbottom resources are documented during the post-construction monitoring event.

3.4.3 Actions Required if Impacts to Resources are Observed

For each instance in which project related impacts to hardbottom resources are detected, the FDEP JCP Compliance Officer shall immediately (within 24 hours) be notified. A detailed Impact Assessment shall also be conducted within 48 hours of detecting impacts. Impact assessments shall include a delineation (using survey-grade DGPS) of all areas in which hardbottom resources have been damaged, injured, buried, or stressed. The condition of impacted benthic organisms shall be assessed and documented (video and photographs) and organisms shall be triaged, if possible. Unless a time extension is requested by the Permittee and granted in writing by the FDEP, results of impact assessment surveys (data and report) shall be submitted to the FDEP JCP Compliance Officer within 10 days of completing the assessment (**Section 7.3.4**). The Permittee shall use the results of the impact assessment to develop a Corrective Action Plan. Unless a time extension is requested by the Permittee and granted in writing by the FDEP, the corrective action plan shall be submitted to the FDEP within 30 days of completing the impact assessment. Once agreed to by the FDEP, damage to hardbottom resources shall be remediated by the Permittee as per the approved corrective action plan. Impacted areas (even those subsequently remediated) will require monitoring and may require mitigation and/or restoration, depending on the outcome of remediation and the scale of the impact(s). Pre-impact baseline video (pre-pumping for pipeline corridors) of areas where (or near where) impacts have occurred will be visually analyzed to aid the FDEP in UMAM analysis.

4.0 MITIGATIVE ARTIFICIAL REEF MONITORING

A total of 0.97 net acres of mitigative artificial reef shall be constructed to offset 0.57 acres of direct impacts to nearshore hardbottom within the permitted ETOF and pipeline corridors. Post-construction monitoring is required to document the constructed net acreage of hardbottom and the community that forms on the artificial reef. These data provide the basis for determining the success of the mitigation reef in providing the net acreage of hardbottom required to offset project impacts and in developing a hardbottom community similar (structure, composition, and function) to that of the reference community. Hardbottom seaward of the permitted ETOF but within the project area shall serve as the reference area for the mitigative artificial reef (**Section 4.2**). The reference community shall be documented along 16 permanent project area transects prior to construction during the baseline monitoring event (**Table 1**). Mitigative artificial reef monitoring will include physical surveys (starting immediately after construction [initial as built survey])

(**Section 4.1**) and annual post-construction biological monitoring events (starting the year after construction) (**Section 4.3**). Mitigation success criteria are provided in **Section 4.4**.

4.1 Mitigative Reef Physical Monitoring

A physical monitoring event shall be conducted within six (6) months of construction (as-built survey) of the mitigative artificial reef and another shall be conducted three years after construction (Year 3 post-construction). These monitoring events will provide the FDEP with reasonable assurance that the artificial reef has met its physical (net acreage) success criterion (see Criterion No. 3 in **Section 4.4**). The aim of each physical monitoring event is to document the net acreage of artificial hardbottom present within the artificial reef footprint (gross delineated acreage). The estimated net acreage (see **Section 4.1.3** below) documented during each physical survey shall be compared to the mitigation acreage arrived at by UMAM analysis to determine if the net acreage required to offset impacts has been constructed and is still present (exposed) (see **Section 4.4**). For each physical monitoring event, net acreage shall be calculated as the product of the gross acreage (area within the footprint of each artificial reef) and the percentage of hardbottom within the reef footprint (**Section 4.1.3**). Gross hardbottom acreage shall be measured by delineation of the artificial reef boundary (either in situ or remotely) (**Section 4.1.1**). The percentage of hardbottom within the reef footprint shall be based on the measured ratio of hardbottom to gaps (gaps/sand between boulders) as determined by line intercept surveys along transects (**Section 4.1.2**). Gross and net acreage estimates along with all raw data used in calculations (i.e., results of in situ artificial reef delineation and line-intercept surveys) shall be provided to the FDEP following each physical survey.

4.1.1 Mitigative Artificial Reef Boundary Delineation

The boundary of the mitigative artificial reef shall be delineated (mapped) during each physical monitoring event. Delineation may either be conducted in situ using divers or remotely using multibeam sonar. For in situ boundary delineation, divers will swim the edge (perimeter) of each mitigative reef module while towing a buoy equipped with a DGPS antenna attached by a cable to a Hypack navigation software system onboard a survey vessel or equivalent wireless radio system interfaced with GIS. If delineation is conducted remotely using multibeam sonar, then the same methods employed during the initial (as-built) survey shall be employed in all future mitigative artificial reef surveys for gross acreage (i.e., following collection of multibeam sonar data, staff shall use the same methods (level of precision) to delineate the artificial reef). Once the reef is delineated (either by divers or using multibeam sonar), the total acreage of the delineated area (= gross acreage) shall be determined and reported.

4.1.2 Line-Intercept Surveys Along Temporary Transects

A single layer of limestone boulders placed on the seafloor to form an artificial reef will, due to the size and shape of the boulders used, the configuration in which the boulders are placed, and the tightness with which the boulders are packed together, contain gaps/voids of varying sizes. As such, the net hardbottom acreage of an artificial reef will inevitably be less than its gross delineated acreage. The aim of the line-intercept survey along transects is to measure the percentage of hardbottom present within the artificial reef footprint (gross delineated acreage).

Line-intercept surveys conducted during each physical monitoring event shall employ a total of 18 temporary, randomly placed 30 m long transects. Prior to each physical monitoring event, the position of each temporary transect shall be determined by randomly generating a start point and degree heading. Once start points and degree headings are set, monitoring team divers will use the GPS coordinates to position each temporary transect when conducting the line-intercept survey. Physical monitoring events are the only time (monitoring events) in which temporary artificial reef transects will be surveyed.

During the survey, a diver will swim the length of each transect and, using a plumb bob, note the location along the transect tape, and measure the linear extent, of artificial hardbottom (boulders) and all gaps between boulders. Following data collection, the percentage of each transect line accounted for by boulders (hardbottom) and by gaps/sand will be calculated for each transect.

4.1.3 Net Hardbottom Acreage Calculation

For each physical monitoring event, net acreage shall be calculated as the product of the gross acreage (area within the footprint of each artificial reef) and the mean percentage of hardbottom within the reef footprint. For each monitoring event, the percentage of linear area covered by artificial hardbottom (boulders) along each transect (as documented during the line-intercept survey [**Section 4.1.2**]) shall be averaged across all 18 temporary transects to determine the mean percentage of hardbottom. This mean value shall be multiplied by the gross acreage (in situ delineated reef acreage [**Section 4.1.1**]) of the reef to arrive at an estimate of net hardbottom acreage. The estimated net acreage shall be compared to that required in Mitigation Success Criterion No. 3 (see **Section 4.4**).

4.2 Natural Hardbottom Reference Community Characterization

Natural hardbottom seaward of the ETOF and adjacent to the project area, between R-100 and R-113 in St. Lucie County (**Table 1** and **Figures 3** and **4**), shall be monitored (quantitatively characterized) prior to the initial fill placement event. The community documented within quadrats along transects during this baseline nearshore hardbottom monitoring event (conducted in August of 2012) shall serve as the reference community for the mitigative artificial reef. See **Section 2.1.2.3.3** for quadrat sampling methods.

4.3 Mitigative Reef Biological Monitoring

Biological monitoring is aimed at documenting the community that forms on the artificial reef so that the success of the mitigation in developing a hardbottom community similar (structure, composition, and function) to that of the reference community can be determined. Annual post-construction biological surveys (years 1-3 post-construction) of permanent quadrats along permanent transects will document the development of the hardbottom community (structure and composition) on the artificial reef over time

4.3.1 Establishment of Permanent Artificial Reef Transects and Quadrats

A total of 12 permanent 10 m long transects shall be permanently installed at the start of the year 1 post-construction monitoring event and used to monitor the mitigation reef constructed to offset impacts. Each transect shall be assigned to a randomly selected reef "cell" (module) prior to installation. Ten (10) permanent transects will be established at the artificial reef between R-70 – R-73, the other two transects will be established at the artificial at R-90. A total of 5 permanent quadrats shall be established along each of the 12 permanent transects at the artificial reef during the year 1 post-construction monitoring event. Quadrats (0.25 m^2 each) shall be spaced roughly 2.5 m apart, with the first quadrat placed at roughly the 0 m mark along each transect. Depending on the distribution of boulders within reef modules, some transects may only have 4 quadrats established along their length. Once established, the positions, lengths, and number of transects will remain the same throughout the course of monitoring. GPS coordinates for the start, middle, and end of each permanent transect shall be recorded and archived.

4.3.2 Monitoring Permanent Artificial Reef Quadrats and Transects

4.3.2.1 Monitoring Permanent Artificial Reef Quadrats

The benthic mitigation reef community will be characterized quantitatively during three (3) annual monitoring events (years 1-3 post-construction) by surveying the assemblage within each permanent quadrat along each permanent transect. The use of permanent quadrats ensures the same habitat and assemblages (same location and area) are sampled each monitoring event so that changes in the cover of organisms can be documented. The survey protocol is the same as that employed in the baseline nearshore hardbottom monitoring event (**Section 2.1.2.3.3**).

4.3.2.2 Video Recording Along Permanent Artificial Reef Transects

Video survey data collected as part of mitigation reef monitoring shall function as an archival dataset that can be used for general reference purposes and to document changes in the benthic community along transects over time. As such, video recordings must be of a quality sufficient to allow for post-collection quantitative image analysis using point count procedures. Therefore, surveys shall be conducted using a high-definition digital video camera in a waterproof housing.

Video surveys will be conducted along each permanent artificial reef transect during each monitoring event. Video of the artificial reef along each transect will progress no faster than 5 meters per minute. A convergent laser guidance system will be used during the survey to precisely maintain the height of the camera at 40 cm above the seafloor. The transect line will be clearly visible in all video so that locations may be accurately referenced. A 360° panoramic view at an angle of roughly 30° to the horizon will be recorded at both the beginning and end of each transect from an elevation of roughly 1 m above the seafloor. The panoramic recording shall take at least 30 seconds to conduct (full revolution). At the beginning and end of each transect, a standard underwater display will also be recorded and integrated directly onto the digital video track. The standard display will report: 1) the permit number (standard DEP/JCP project number) (e.g., 0555555-001-JC); 2) the artificial reef transect number/identifier; 3) the survey date (e.g., 06/25/2018); 4) the water depth in meters for both the beginning (transect meter 0) and end (final meter) of the transect (e.g., start depth = 2 m, end depth = 4.5 m); and 5) any pertinent notes (e.g., poor visibility, large swell, etc.). Video data (files) will be supplied to FDEP during raw data submittal.

4.4 Mitigation Success Criteria

The FDEP typically considers mitigation to be successful, and the reef to have performed adequately, when: the net acreage meets that required to offset the acreage of impacts; and, the benthic community of the artificial reef is comparable to that of the reference community. The following success and performance criteria shall be met:

1) 90% of species found in the impact site shall be present in the mitigation site by the time of the completion of the monitoring period;

2) Percent cover by the major groups of organisms (functional groups) in the mitigation site shall be no less than it was in the impact site; and

3) A line-intercept survey shall demonstrate that net amount of reef versus sand did not change from the time of construction (+ 5% from results of initial survey).

If, after three (3) years of monitoring, all success criteria have not been achieved, then the Permittee and/or authorized agent shall meet with the FDEP to discuss the performance of the mitigation reef and a path forward to complete the mitigation requirement.

5.0 IN-WATER SEA TURTLE MONITORING

Monitoring will be conducted for one-year (months 0 to 12) following each construction event. Starting with the immediate post construction monitoring event, monitoring will be conducted quarterly when sea conditions are ideal for performing surveys. This will yield a total of four (4) monitoring events per nourishment event.

In-water sea turtle surveys will be conducted from a boat equipped with an elevated observation platform and a GPS navigational system to allow constant monitoring of speed and location. Surveys will only be conducted during periods of calm seas (2.5 feet or less), with no whitecaps or breaking waves, other than wave break on the beach. The boat will traverse pre-established shore-parallel transects of fixed length (**Figure 6**, EAI 2009) at slow speed (4.0 knots or less). Because the North Segment (i.e. R-87.7 to R-90.3) was deleted from the Project, the previous Control 1 transect will be deleted and the North transect will serve as the second control transect. Two observers will be positioned on the platform and will make observations at an eye level of approximately 12 feet above the water; one will monitor the port side and the other the starboard side. Two passes will be made along each transect with observers changing sides between the first and second passes.

When a turtle is sighted, the position of the vessel at the time of the sighting will be recorded by GPS. The position of the turtle will be mapped based on the position of the boat and the observer's estimate of the distance and bearing to the turtle. In addition to recording the position of the turtle, the observer will also record the time the turtle was first observed, the species, its relative size, and whether it was at the surface or submerged.

Of the two passes made along each transect during each monitoring event, the pass with the highest number of sea turtle sightings will be used for estimating density of turtles along that transect. A belt extending 50 meters to either side of the survey vessel over the length of each transect will be used for estimating density. Post-construction density data will be used for documenting natural spatial and temporal variability of turtles within the Study Area and will be used in comparison with baseline monitoring data to assess potential project effects. In-water turtle monitoring for the mitigation reef is detailed in the Mitigation Plan (CTC 2012).

Figure 6: Location of in-water sea turtle monitoring transects, as established by Ecological Associates, Inc. (EAI) in 2008 and 2009 (from EAI 2009).

6.0 MONITORING SCHEDULE AND TEAM REQUIREMENTS

6.1 Monitoring Schedule

Nearshore hardbottom monitoring for the initial beach fill placement under this permit shall include pre- and post-construction monitoring (**Table 3**). Each subsequent nourishment event shall initiate a complete round of post-construction nearshore hardbottom and in-water turtle monitoring. Pipeline corridor surveys shall be conducted when the borrow area is used as a sand source for a fill placement event.

Table 3. Monitoring summary and schedule.

Section	Survey	Survey Type	Survey Period and Number of Events	Deliverables
2.1 Nearshore Hardbottom	26 Permanent Transects outside ETOF (N=16 in project area and N=10 in Reference Areas); Permanent Quadrats (0.25 m^2 each)	Line-Intercept	*Pre-Construction* (N=1): Once prior to initial fill placement (Baseline)	Excel spreadsheets, PDF of field sheets
		Interval Sediment Depth		
		Quadrat Sampling		
		Qualitative Video	*Post-Construction* (N=4 per fill placement event): Immediately (within 6 mo.) and annually for 3 years (1, 2, and 3)	Video
	Hardbottom Edge	In situ Diver Delineation of Nearshore Edge		Shapefiles
2.2 Sonar	Hardbottom Sonar Survey	Hardbottom Delineation and Diver Verification		Shapefiles / Geodatabase
3.0 Pipeline Corridors*	Pre-Construction Corridor Area Survey	Hardbottom Mapping and Characterization	*Pre-construction* (N=1 prior to each placement event)	Video, PDF of field sheets, and Shapefiles
4.0 Mitigative Artificial Reef	Reference Community	Sampling quadrats along permanent NSHB transects	*Pre-Construction of nourishment:* (N=1 full survey prior to first fill placement event)	Excel spreadsheets, PDF of field sheets
	Physical Monitoring	Artificial Reef Boundary Delineation	*Post-Construction of the mitigative reef:* Immediately following mitigative reef construction and year 3 post-construction	Shapefiles and calculations
		Line-intercept along 18 temporary transects [30 m long each]		Excel spreadsheet, PDF of field sheets
	Biological Monitoring	Sampling of 0.25 m^2 Quadrats (along 12 permanent transects [10 m long each]; N=5 quadrats / transect)	*Post-Construction of the mitigative reef* (N=3): [Years 1, 2, and 3] monitoring events)	Excel spreadsheet, PDF of field sheets
		Video (full length)		Video
5.0 In-water Sea Turtle	Permanent Transects	Counts from Vessel	*Post-Construction* (N=4 per fill placement event): Quarterly from 0 to 12 mo. after construction	Excel spreadsheets, PDF of field sheets

* Only required when an offshore borrow area(s) will be/is used as a sand source for construction event.

6.2 Monitoring Team Requirements

The names and qualifications of the staff performing the biological monitoring shall be submitted by the Permittee or their Agent to the FDEP for approval. Written agency approval of personnel will be required prior to proceeding with the yearly monitoring. Biological monitoring surveys shall be conducted by staff that are certified SCUBA divers with previous experience in monitoring hardbottom communities and with scientific knowledge of local benthic marine ecosystems and flora and fauna. In addition to this, all in-water crew members responsible for in situ quadrat data collection shall have a BS degree or higher in the study of marine biology or a comparable field and shall have scientific knowledge of local benthic marine hardbottom habitats and their flora and fauna. These crew members shall also participate in cross training to verify correct species identification and survey practices as Quality Assurance/Quality Control (QA/QC) procedures at the beginning of each monitoring event. QA/QC results shall reflect consistency of 90% for percent cover and identification of functional groups between observers.

7.0 REPORTING

7.1 Notification of Commencement, Progress, and Completion of Work

Commencement dates of surveys shall be reported via email to the FDEP JCP Compliance Officer (JCPCompliance@dep.state.fl) and to FDEP staff in the Beaches, Inlets, and Ports program roughly seven (7) days prior to the start of monitoring and the day that monitoring begins. Brief monitoring progress reports shall be submitted (emailed) weekly to the JCP Compliance Officer until completion of the monitoring event. As soon as monitoring activities have ended, the JCP compliance officer shall be notified that the monitoring event has been completed.

7.2 Nearshore Hardbottom Monitoring Data and Report Submissions

7.2.1 Pre-Construction (Baseline) Monitoring Data

A single pre-construction (baseline) nearshore hardbottom monitoring event shall be conducted prior to the initial fill placement event (**Section 2.1**). This monitoring event also serves as the reference community survey for the mitigative artificial reef (**Section 4.2** and see **Section 7.4.1**). <u>At least 30 days prior to construction,</u> pre-construction monitoring data shall be submitted directly and concurrently by the monitoring firm to the FDEP JCP Compliance Officer, the Permittee, and the Agent (e.g., on portable hard drives or via an FTP site). All data submitted shall be provided in standard formats, as specified below. All transect monitoring data submitted shall have been checked against field datasheets and corrected (if necessary) to ensure accuracy. Raw data provided shall consist of the following, each of which are described below: video and photographs, hardbottom edge survey data, raw transect survey data, and field datasheets. A side scan sonar survey shall be conducted prior to the second fill placement event conducted under the permit (**Section 2.2**). Pre-construction side scan sonar monitoring data shall be submitted to the FDEP as described above and <u>at least 30 days prior to construction.</u>

7.2.1.1 Video and Photographs

Qualitative digital video and any digital photographs shall appear in separate folders. Main folders and subfolders shall be identified by descriptive names, so data may be easily differentiated (e.g., by transect).

7.2.1.2 Hardbottom Edge Survey Data

Hardbottom edge data shall be supplied as a collection of shapefiles (e.g., as an ESRI file geodatabase). Lines or polygons shall represent the in situ mapped landward edge of hardbottom or landward hardbottom patches for data obtained from each survey. Depending on the distribution of hardbottom, these data may be depicted as a single line representing the nearshore edge or as polygons representing hardbottom patches.

Hardbottom edge data shall have attributes indicating the portion of each line or polygon representing hardbottom. If sand patches greater than 5 m are crossed during the edge survey, these portions of lines/polygons shall, as attributes, be indicated as sand. Lines/polygons representing the permitted ETOF shall also be provided with the collection of shapefiles.

7.2.1.3 Transect and Quadrat Survey Data

Interval sediment depth measurements, line-intercept data, and BEAMR quadrat data collected along transects shall be supplied in Excel format. Each Excel workbook submitted shall include a descriptive name, so data may be easily differentiated by area.

7.2.1.4 Field Datasheets and Survey Logs

Copies (photographs or scans) of field datasheets shall be submitted in pdf format.

7.2.1.5 Hardbottom Side Scan Sonar Survey Data

Hardbottom delineations based on side scan sonar shall be supplied as a geodatabase. Raw sonar survey records shall also be provided.

7.2.2 Post-Construction Monitoring Data Submissions

<u>Within 45 days of completing each required post-construction monitoring event</u>, all raw data shall be submitted directly and concurrently by the monitoring firm to the FDEP JCP Compliance

Officer, the Permittee, and the Agent (e.g., on portable hard drives or via an FTP site). Raw data provided to the FDEP shall consist of the following, each of which are described in **Sections 7.2.1.1 – 7.2.1.5**: video and photographs, hardbottom edge survey data, raw transect survey data, field datasheets, and hardbottom delineations and associated side scan sonar survey data. All data submitted shall be provided in standard formats and transect monitoring data submitted shall have been checked against field datasheets and corrected (if necessary) to ensure accuracy.

7.2.3 Post-Construction Monitoring Report Submissions

Within 90 days of completing each required post-construction monitoring event, a written monitoring report shall be submitted directly and concurrently by the monitoring firm to the FDEP JCP Compliance Officer, the Permittee, the Agent and to the FFWCC, USACE, USFWS and/or NMFS. Each monitoring report shall clearly describe methods used in monitoring and data analysis and explain any deviations from the monitoring plan or conditions of permit. Reports shall also provide results in appropriate graphical, tabular and text formats. Monitoring reports are to be cumulative; thus, data (in the form of summary tables and figures) from all previous monitoring efforts shall be included in each report, in an updated fashion. Not all data sets will be analyzed and compared statistically. Temporal comparisons by way of univariate and multivariate tests shall be confined to data collected during the most recent monitoring event (current survey) and the baseline survey. Statistical tests will not be used to compare results between different postconstruction monitoring events. Noteworthy explanatory observations and other ancillary information shall be provided at the end of the report in an appendix.

7.3 Pipeline Corridor Survey Data and Reports

Pre-construction pipeline corridor surveys are required each construction event (fill placement) in which the borrow area is used as a sand source (**Section 3.0**).

7.3.1 Pre-Construction Corridor Survey Data Submissions

At least 45 days prior to any and all construction activities, all pre-construction pipeline corridor survey data (**Section 3.0**) shall be submitted directly and concurrently by the monitoring firm to the FDEP JCP Compliance Officer, the Permittee, and the Agent (e.g., on portable hard drives or via an FTP site). Data submitted shall include the following:

- A list of all pipeline corridors and their locations (shapefiles)
- Data and video collected during all diver surveys
- Hardbottom mapping data (shapefiles)

Diver survey data shall be provided as videos, in excel workbooks, and as raw copies of field datasheets and survey logs (scanned in color and provided in pdf format). All survey data submitted shall have been checked against field datasheets and corrected (if necessary) to ensure accuracy.

Hardbottom mapping results for each pipeline corridor shall be provided as a collection of shapefiles (e.g., as a file geodatabase). For shapefiles, lines or polygons shall represent the in situ mapped edges of hardbottom patches/features.

7.3.2 Pre-Construction Corridor Survey Report Submissions

At least 30 days prior to any and all construction activities, a written pre-construction pipeline corridor survey report shall be submitted directly and concurrently by the monitoring firm to the FDEP JCP Compliance Officer, the Permittee, the Agent. The report shall clearly describe methods used in surveys and data analysis and provide results of each pipeline corridor survey. The report shall identify corridor survey areas where: A) hardbottom resources are currently absent; B) hardbottom resources are currently present but the Permittee will avoid resources during construction; and C) hardbottom resources are currently present and the Permittee will employ minimization measures to limit impacts to resources. The type of minimization measures (e.g., collars or risers) and the specific locations in which they will be employed shall also be included in the report. Note that impacts to hardbottom resources due to construction activities are required be avoided to the greatest extent practicable.

7.3.3 Post-Placement Pre-Pumping Pipeline Survey Data Submissions

At least 72 hours prior to the intended start or actual start of pumping, post-placement, pre-pumping pipeline survey data for each placed pipeline required to be surveyed (**Section 3.3**) shall be submitted to the FDEP JCP Compliance Officer in electronic format (e.g., on a single portable hard drive or via an FTP site or email). Survey mapping results shall be provided as a collection of shapefiles (e.g., as an ESRI file geodatabase). For shapefiles, a line or lines shall represent the in situ mapped position of the placed pipeline. Lines or polygons shall also be used to represent the in situ mapped edges of hardbottom patches/features documented during the current pre-construction hardbottom mapping effort (**Section 3.1**) within corridors. Hardbottom within 25 m to either side of the placed pipeline shall be highlighted.

7.3.4 Pipeline Corridor Impact Assessment Data and Report Submissions

For each instance in which impacts to hardbottom resources are documented within pipeline corridors during or following construction, results (data and report) of the impact assessment (**Section 3.4.3**) shall be submitted to the FDEP JCP Compliance Officer within 10 days of completing the assessment, unless a time extension is requested by the Permittee and granted in writing by the FDEP. In combination, the data and report submitted for each impact assessment shall provide all information necessary for the FDEP to calculate the amount of compensatory mitigation that may be required to offset unpermitted impacts using the Uniform Mitigation Assessment Method.

Each data set submitted shall include video and photographs collected during the impact assessment as well as the GPS coordinates for locations with impacted hardbottom resources. Mapping results for areas with impacted resources shall also be provided, as a collection of shapefiles (e.g., as an ESRI file geodatabase). For shapefiles, polygons shall represent the in situ delineated edge of each area containing impacted resources. Additionally, baseline monitoring data for impacted hardbottom patches/features (pre-pumping video for pipeline corridors) shall be provided.

Each impact assessment report shall describe, compare, and contrast the pre-impact and post-impact condition of resources within impacted areas. Pre-impact condition shall be based on pre-pumping monitoring of corridors while post-impact condition shall be based on results of the quantitative assessment survey. Documentation of the types of impacts encountered (e.g., physical damage to resources caused by construction equipment or activities) shall also be provided. Additionally, the report shall describe the severity of functional losses documented by comparisons of pre and post impact monitoring and assessment data (e.g., reduction in hardbottom acreage and loss of numbers and types of individuals/colonies).

7.3.5 Post-Construction Pipeline Corridor Data and Report Submissions

<u>Within 45 days of completing post-construction monitoring for each corridor requiring biological monitoring</u> (**Section 3.4**), raw data (video) shall be submitted directly and concurrently by the monitoring firm to the FDEP JCP Compliance Officer, the Permittee, and the Agent and to the FFWCC, USACE, USFWS and/or NMFS (e.g., on portable hard drives or via an FTP site). A brief monitoring report shall also be provided along with each data submittal. Each report shall clearly describe methods used in monitoring and, if occurring, explain any deviations from the monitoring plan or conditions of permit. The report shall also briefly describe results of the monitoring effort and shall indicate whether or not impacts were observed. The report need not describe the extent of impacts or results of impact surveys if impacts were observed, as this information shall be reported separately, as described below.

If impacts to hardbottom resources were documented within corridors during the post-construction monitoring event, then actions required in **Section 3.4.3** shall have been performed and results of the impact assessment shall be reported as specified in **Section 7.3.4**.

7.4 Mitigative Artificial Reef Survey Data and Reports

7.4.1 Mitigative Reef Reference Community Characterization Survey Data

Natural hardbottom outside of the ETOF and adjacent to the project area shall be quantitatively characterized prior to the initial fill placement event during the baseline hardbottom monitoring event (**Section 4.2**). These data serve as the reference community dataset for the mitigative artificial reef and shall be submitted as required in **Section 7.2.1**.

7.4.2 Mitigative Reef Physical Monitoring Data and Report Submission

Within 45 days of completing each physical monitoring event (**Section 4.1**), all raw data shall be submitted directly and concurrently by the monitoring firm to the FDEP JCP Compliance Officer, the Permittee, and the Agent (e.g., on portable hard drives or via an FTP site). Raw data shall consist of artificial reef boundary survey data, raw transect survey data, and field datasheets. Artificial reef boundary (edge) survey data shall be supplied as a collection of shapefiles (e.g., as an ESRI file geodatabase). The in situ mapped boundary of the artificial reef shall be depicted as single or multiple polygons, depending on the distribution of artificial reef boulders. Monitoring transacts shall be depicted as lines. Line-intercept data collected along temporary transects shall be supplied in Excel format. Copies (photographs or scans) of field datasheets shall be submitted in pdf format.

A brief monitoring report shall also be provided along with the data. The report shall provide a description of the survey methods and will compare the calculated net acreage of the mitigation reef to that determined by UMAM and required by the permit (see Criterion No. 3 in **Section 4.4**).

7.4.3 Mitigative Reef Biological Monitoring Data and Report Submissions

7.4.3.1 Mitigative Reef Biological Monitoring Data

Within 45 days of completing each required post-construction biological monitoring event, all raw data shall be submitted directly and concurrently by the monitoring firm to the FDEP JCP Compliance Officer, the Permittee, and the Agent (e.g., on portable hard drives or via an FTP site). Raw data provided to the FDEP shall consist of video and photographs, raw transect survey data, and field datasheets. All data submitted will be provided in standard formats (refer to **Sections 7.2.1.1 - 7.2.1.5** for guidance). All transect monitoring data submitted shall have been checked against field datasheets and corrected (if necessary) to ensure accuracy.

7.4.3.2 Mitigative Reef Biological Monitoring Reports

Within 90 days of completing each required post-construction monitoring event, a written monitoring report shall be submitted directly and concurrently by the monitoring firm to the FDEP JCP Compliance Officer, the Permittee, and the Agent. Along with each monitoring report, the data analyzed to produce the report will also be submitted (e.g., tables used in the analysis of data, tables used to construct figures, and tables and figures provided in the report will be submitted in Excel format). The table entered into Primer and the Primer analysis file will also be submitted.

Each monitoring report will clearly describe methods used in monitoring and data analysis and explain any deviations from the monitoring plan or conditions of permit. Annual biological monitoring reports will be cumulative, thus data (in the form of summary tables and figures) from all previous monitoring efforts will be provided in each report, in an updated fashion. In each report, univariate and multivariate tests will be used to compare the biological community that has developed on the mitigation reef to that characterized in the impact area (reference community). Each report will then indicate and describe the degree to which the artificial reef has succeeded in reaching its success criteria (see Criteria No. 1 and 2 in **Section 4.4**). Noteworthy explanatory observations and other ancillary information will be provided at the end of each report, in an Appendix. Monitoring reports will also include a map showing the location of the artificial reef and permanent monitoring transects overlaid on clear aerial photography.

7.5 In-Water Turtle Monitoring Data and Report Submissions

Within 90 days of completing the final quarterly post-construction survey for each construction event, a report on the distribution and abundance of marine turtles in the vicinity of the nearshore hardbottom in the project area and adjacent, undisturbed control sites (**Section 5.0**) shall be submitted directly and concurrently by the monitoring firm to the FDEP JCP Compliance Officer, the Permittee, the Agent and to the FFWCC, USACE, USFWS and/or NMFS. In addition to the monitoring report, all raw data collected as part of the monitoring effort shall also be provided.

8.0 WORKS CITED

CSA International, Inc. 2010a. Baseline Nearshore Hard Bottom Characterization Survey for the St. Lucie County South Beach Project. 50pp. + appendices.

CSA International, Inc. 2010b. St. Lucie County South Beach Project: Characterization of Hard Bottom Fish Assemblages. 29pp.

Coastal Tech (CTC). 2012. St. Lucie County South Beach and Dune Restoration Project Draft Mitigation Plan. 24 pp. + appendices.

Ecological Associates, Inc. 2009. St. Lucie County South Beach Project: Annual Report of Boat-Based Sea Turtle Surveys December 2008-August 2009. 16pp.